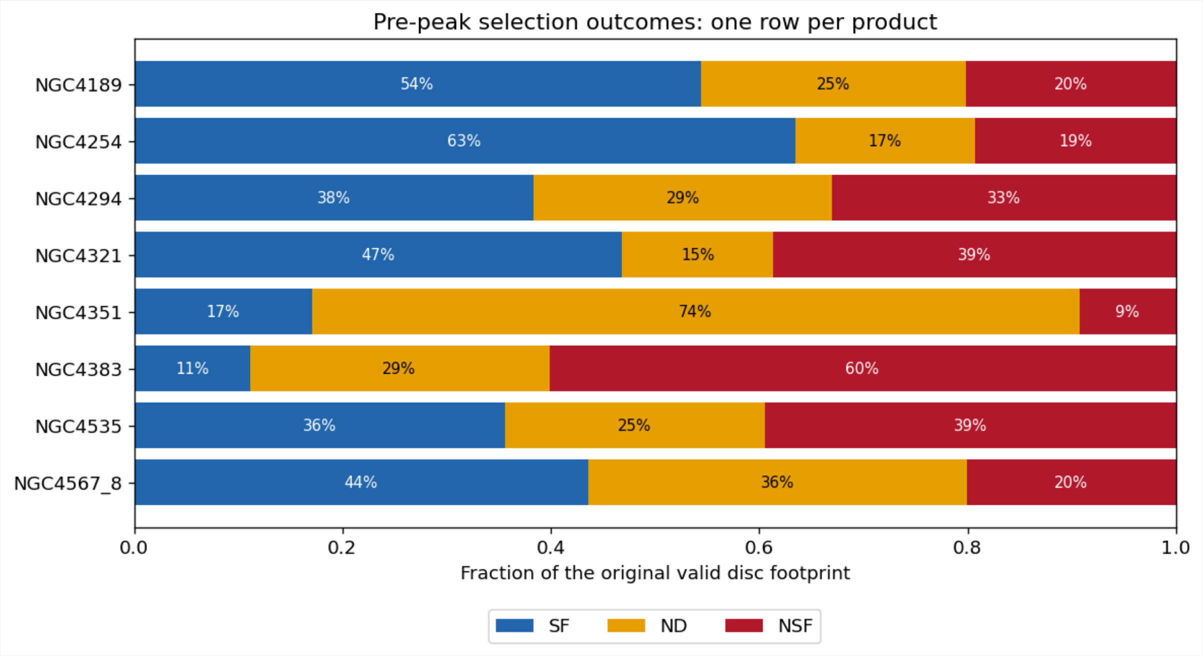
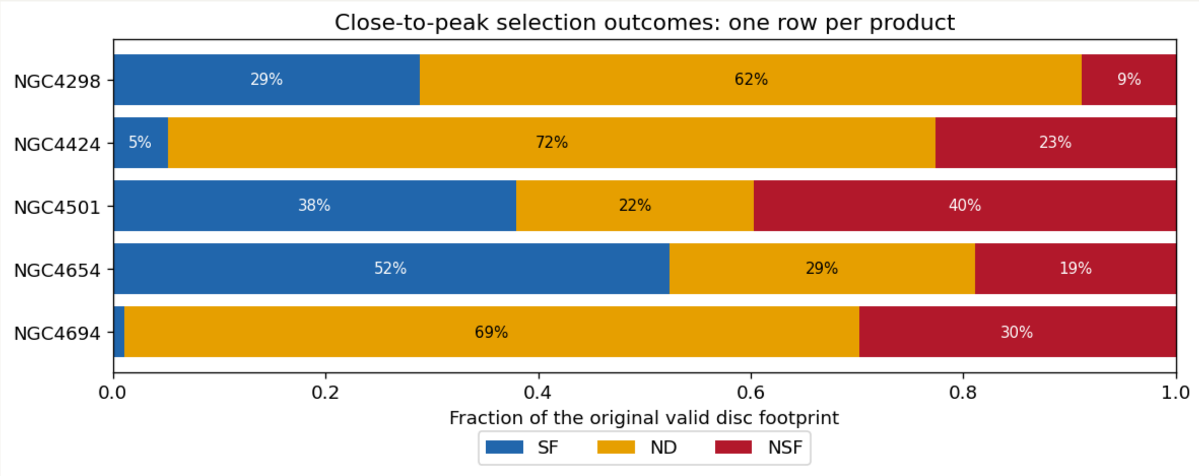
20260901 Close-to-peak and Post-peak rSFMS with SF, NSF and ND

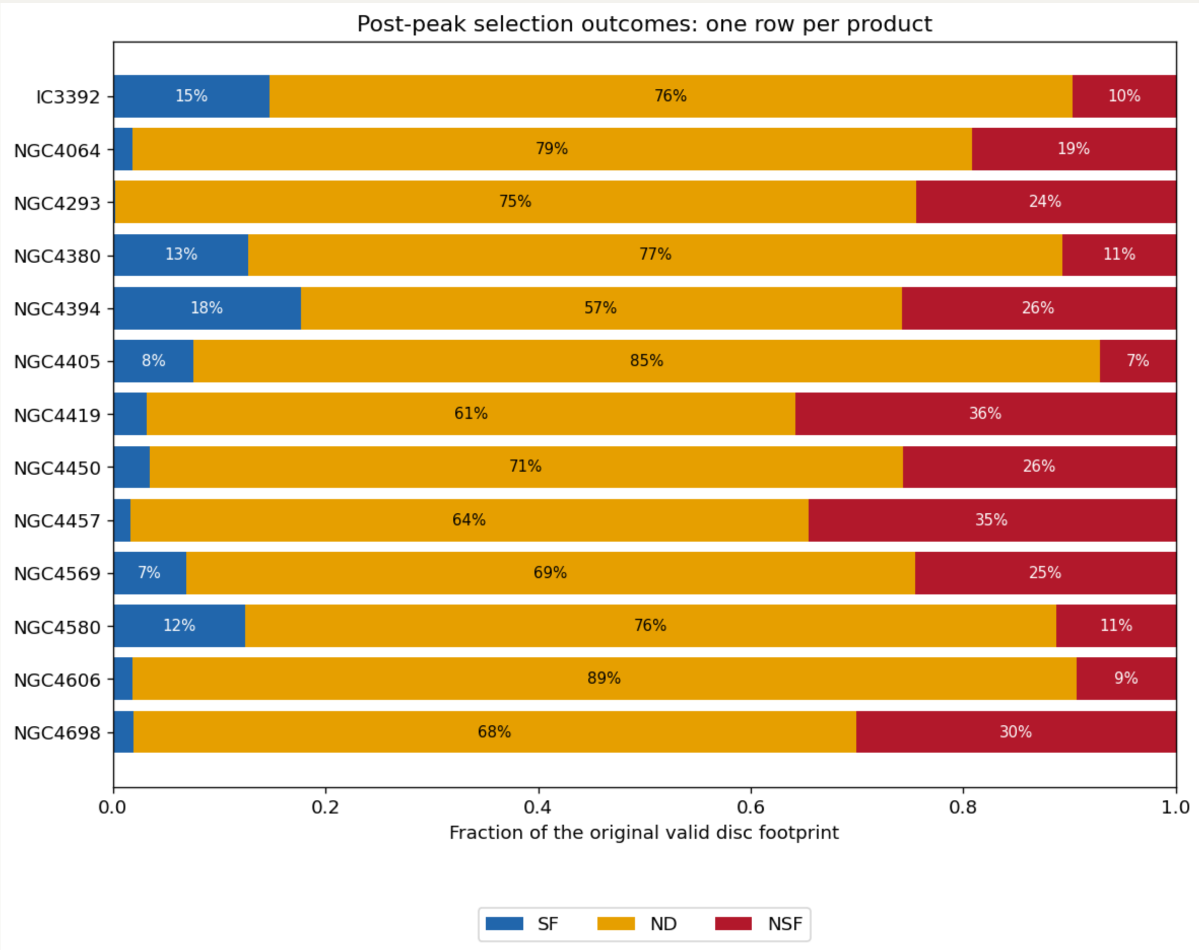
In general, I feel like from pre-peak to close-to-peak to post-peak, the f_{ND} are somewhat similar (even slightly decreasing as we only cover the optial disk in post-peak samples), but the f_{NSF} is gradually increasing.

Across all 3 stages, both SF-only and All samples' stellar mass radial profiles allign very well.

As for SF's, NSF's and ND's distributions in rSFMS, i think we can such idea: if SF and NSF spaxels, especially the SF populations, are quite seperate from the ND spaxels, then we can say for those ND in that galaxy, they are very less likely to be considered as potential SF but not detected spaxels. If in the other case that their occupied parameter spaces are quite connected, especially in y-axis, then we should say we need to treat the ND as upper limit. Also, if ND and NSF are kind of forming another parallel sequence, then those ND must not be SF.



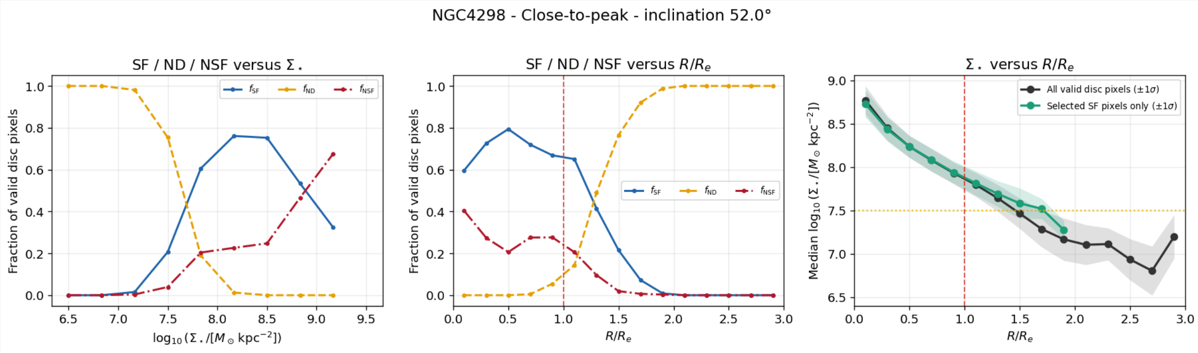


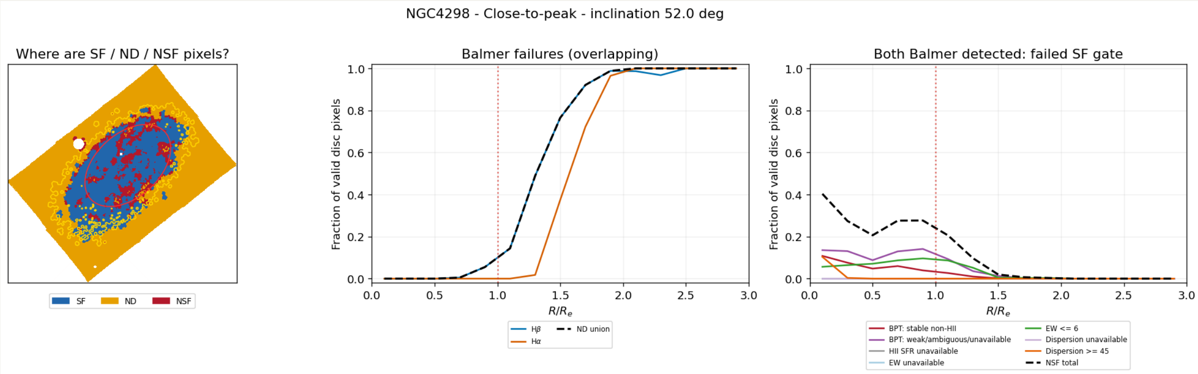


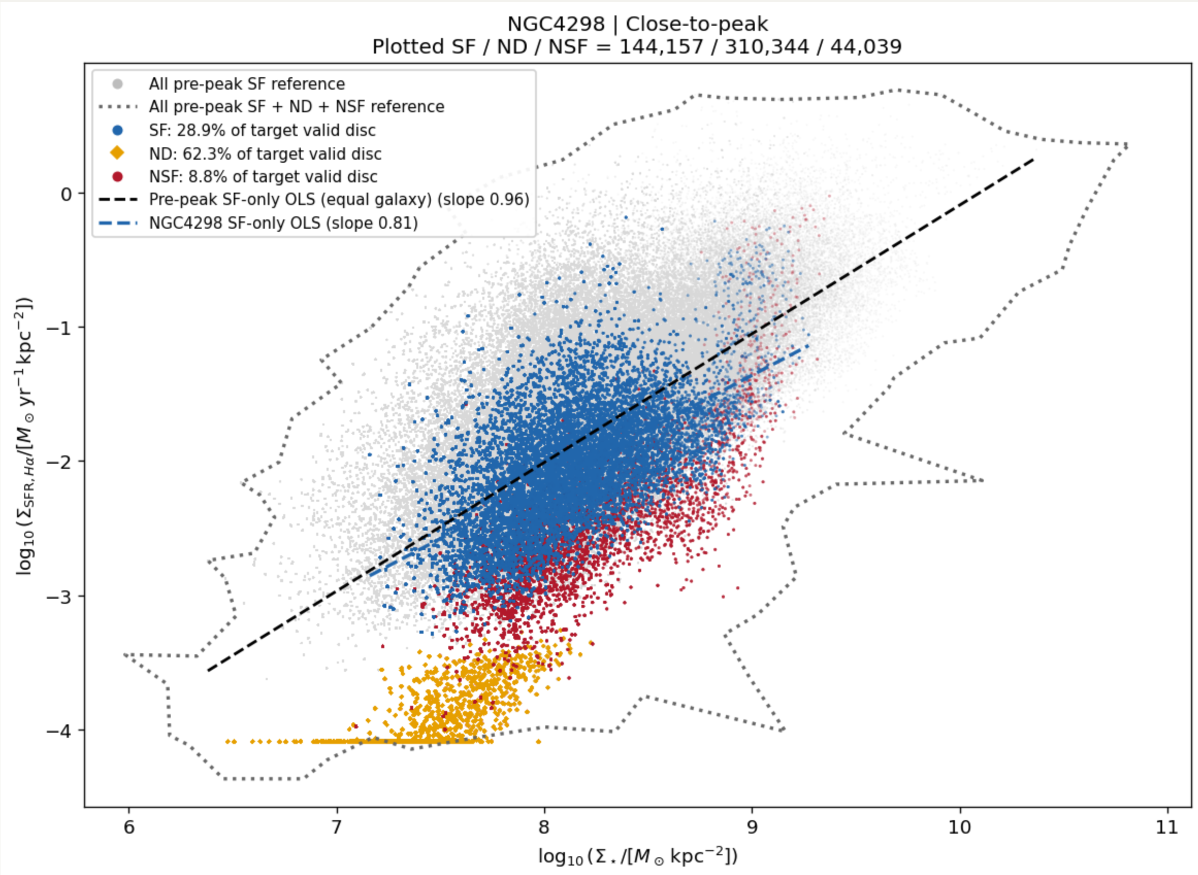
1. Close-to-peak

1.1 NGC4298

While experiencing RPS, this galaxy in rSFMS looks like pre-peak. The SF and NSF spaxels looks quite seperate from ND spaxel in rSFMS. Also its HI gas almost only extend up to its optical disc, so we can say those ND are NSF as well.

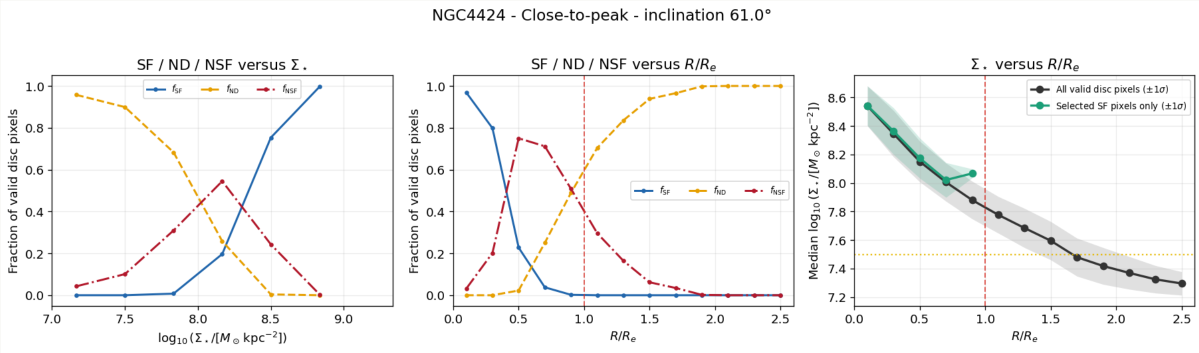


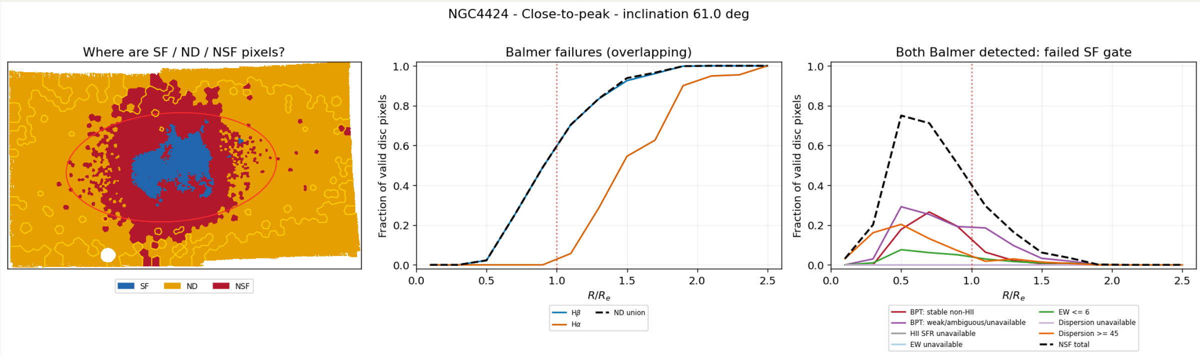


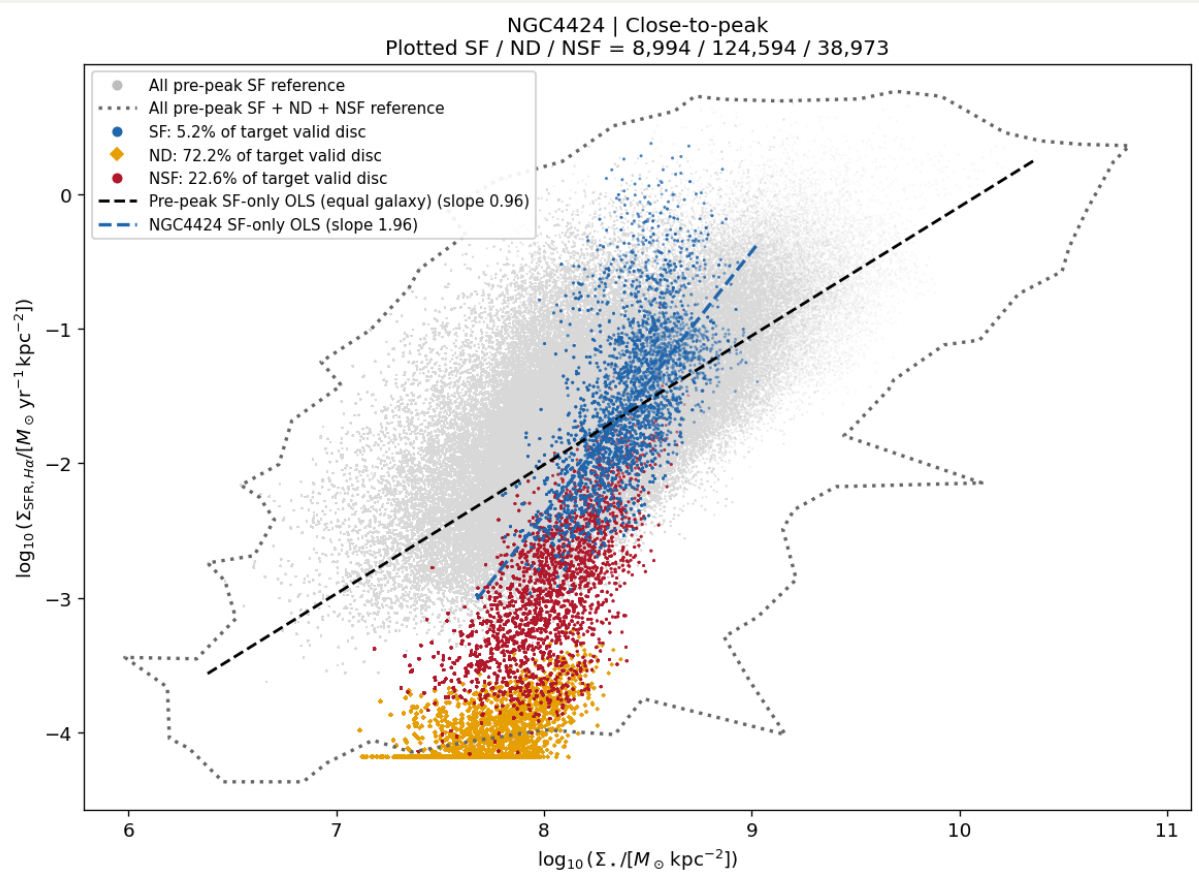


1.2 NGC4424

SF regions are concentrated in the center of the galaxy and completely surrounded by NSF, and then again by ND. In this case i think we can say those ND are truely NSF.

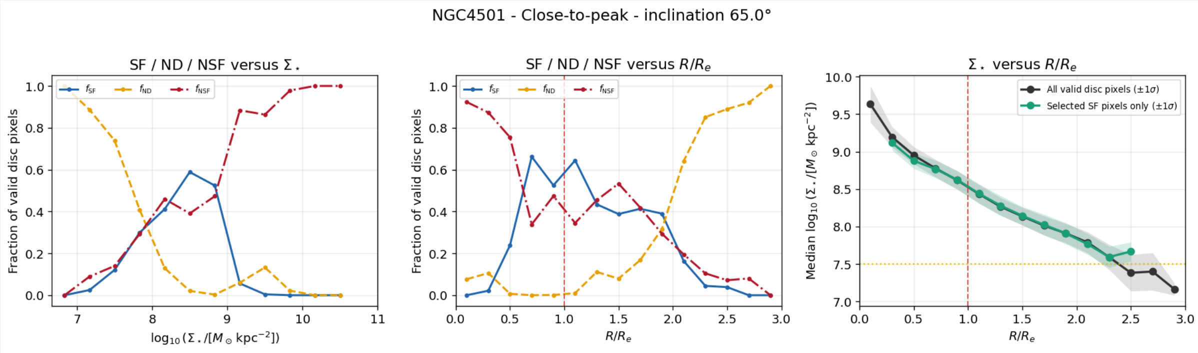


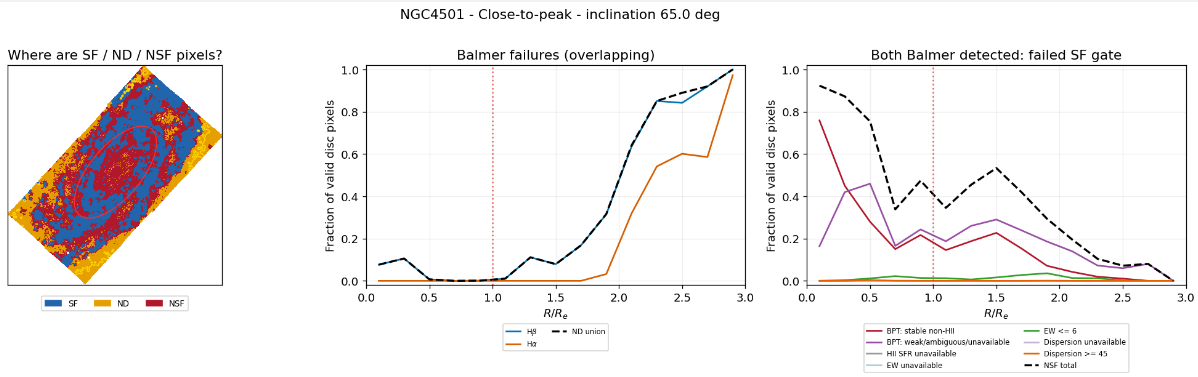


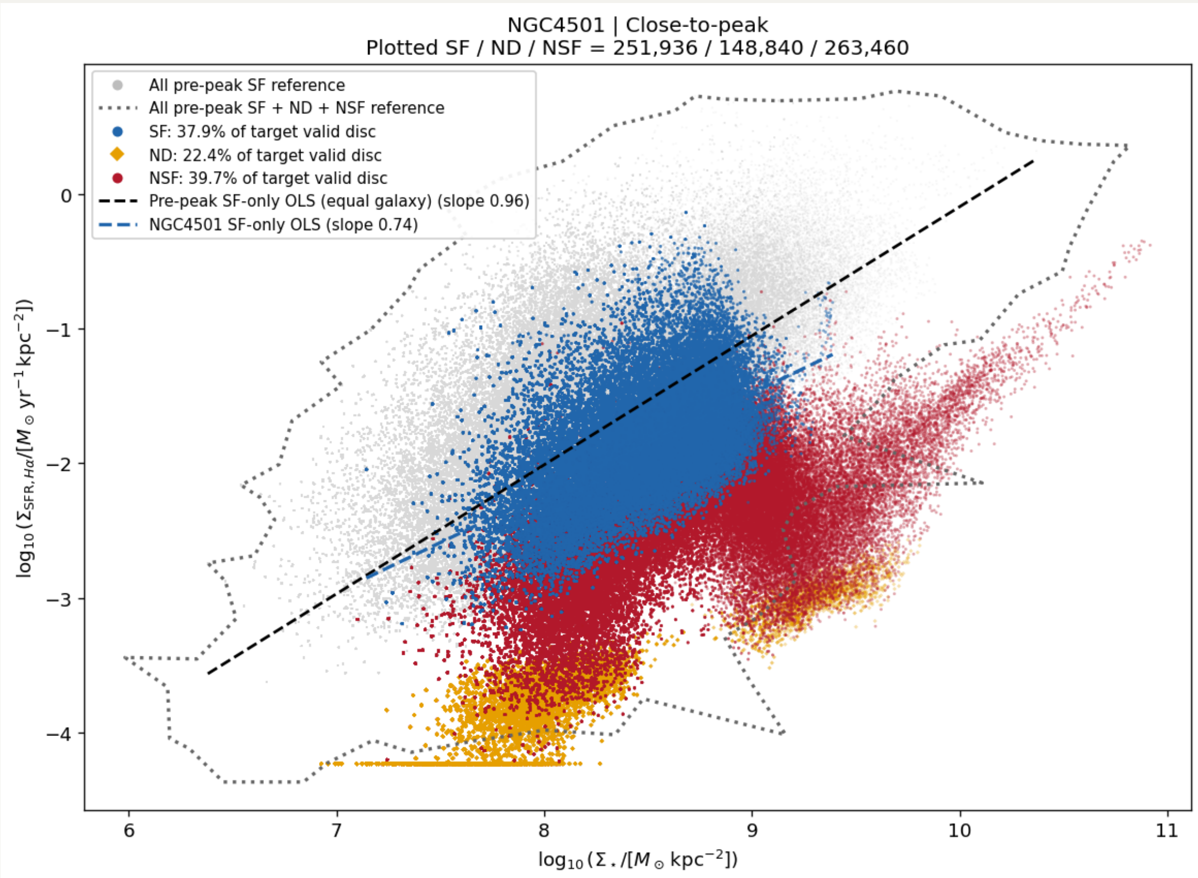


1.3 NGC4501

SF regions are concentrated on spiral arms and those ND should be considered as NSF. Its rSFMS also tends to show that the overall Σ_{SFR} are suppressed.

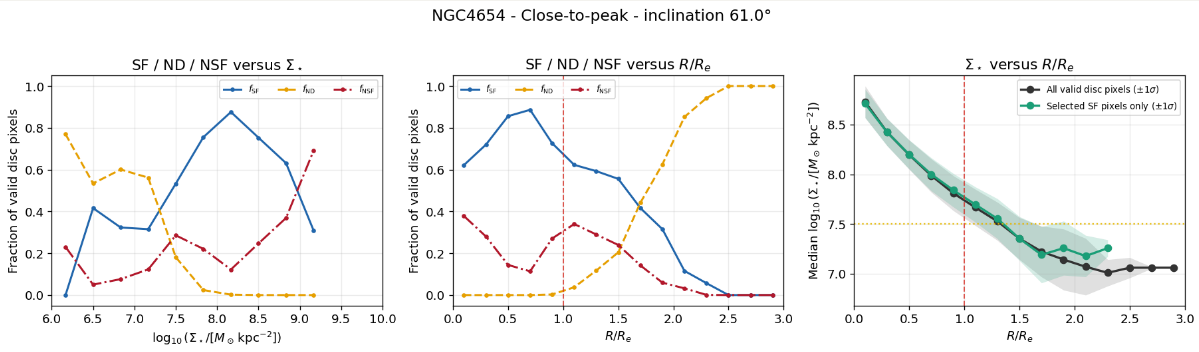

NGC4501 - Close-to-peak - inclination 65.0°
SF / ND / NSF versus Σ.
SF / ND / NSF versus R/Re
Σ. versus R/Re
Fraction of valid disc pixels
Median log10 (Σ./[M⊙ kpc−2])
log10 (Σ./[M⊙ kpc−2])
R/Re
All valid disc pixels (±1σ)
Selected SF pixels only (±1σ)

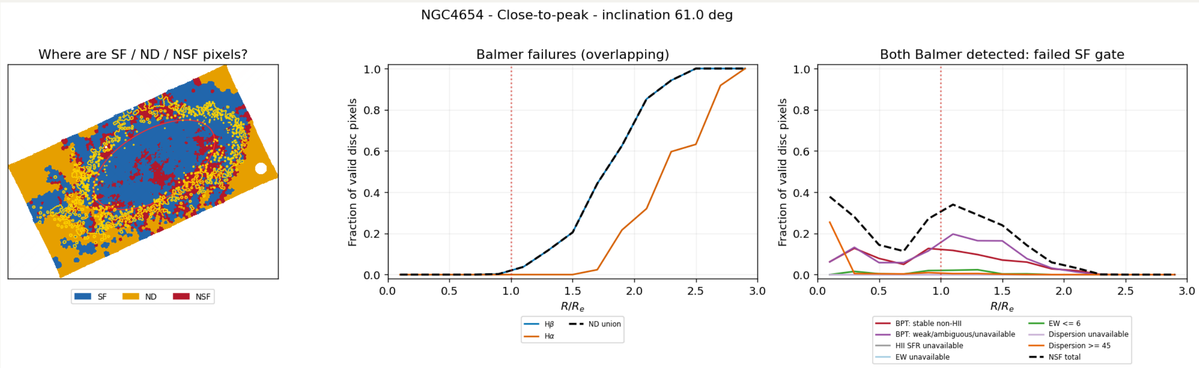

NGC4501 - Close-to-peak - inclination 65.0 deg
Where are SF / ND / NSF pixels?
Balmer failures (overlapping)
Both Balmer detected: failed SF gate
SF
ND
NSF
Fraction of valid disc pixels
R/Re
Hβ
Hα
ND union
BPT: stable non-HII
BPT: weak/ambiguous/unavailable
HII SFR unavailable
EW unavailable
EW <= 6
Dispersion unavailable
Dispersion >= 45
NSF total

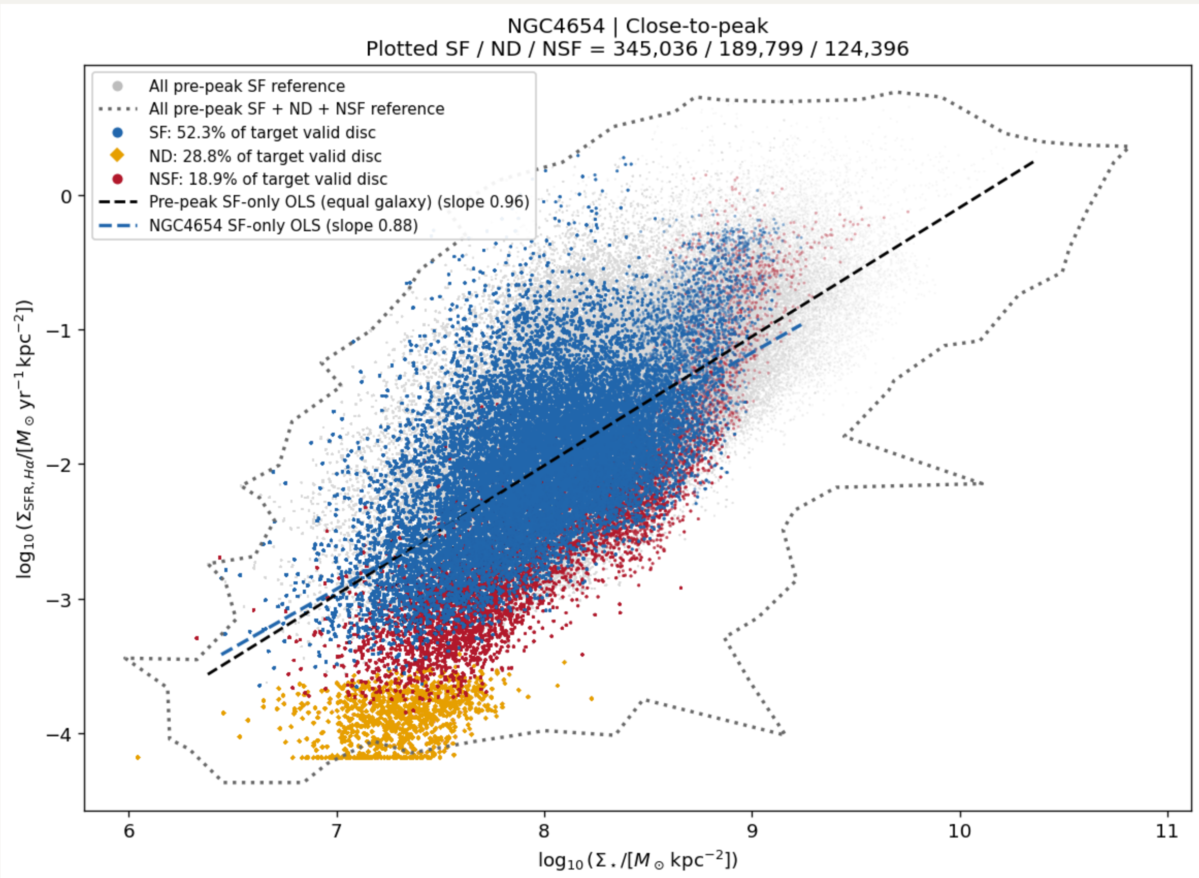

NGC4501 | Close-to-peak
Plotted SF / ND / NSF = 251,936 / 148,840 / 263,460
All pre-peak SF reference
All pre-peak SF + ND + NSF reference
SF: 37.9% of target valid disc
ND: 22.4% of target valid disc
NSF: 39.7% of target valid disc
Pre-peak SF-only OLS (equal galaxy) (slope 0.96)
NGC4501 SF-only OLS (slope 0.74)
log10 (ΣSFR,Hα/[M⊙ yr−1 kpc−2])
log10 (Σ./[M⊙ kpc−2])

1.4 NGC4654

I think this should be best candidate to demonstrate the local SF enhancement (North-West corner, and actually it does). Its rSFMS looks very pre-peak like with some spaxels showing relatively high $\Sigma_{\rm SFR}$.

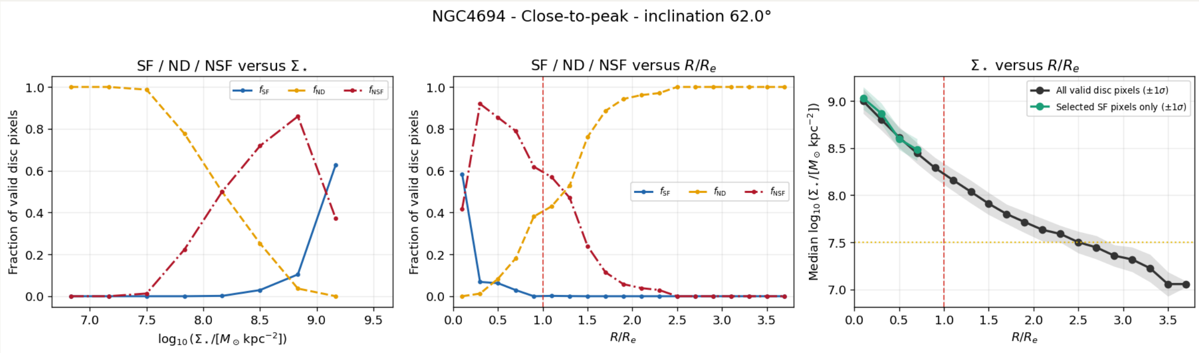


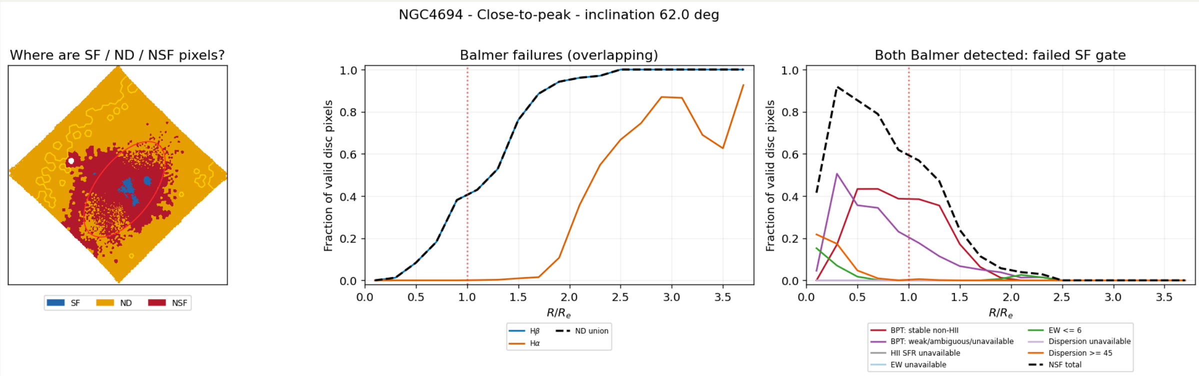


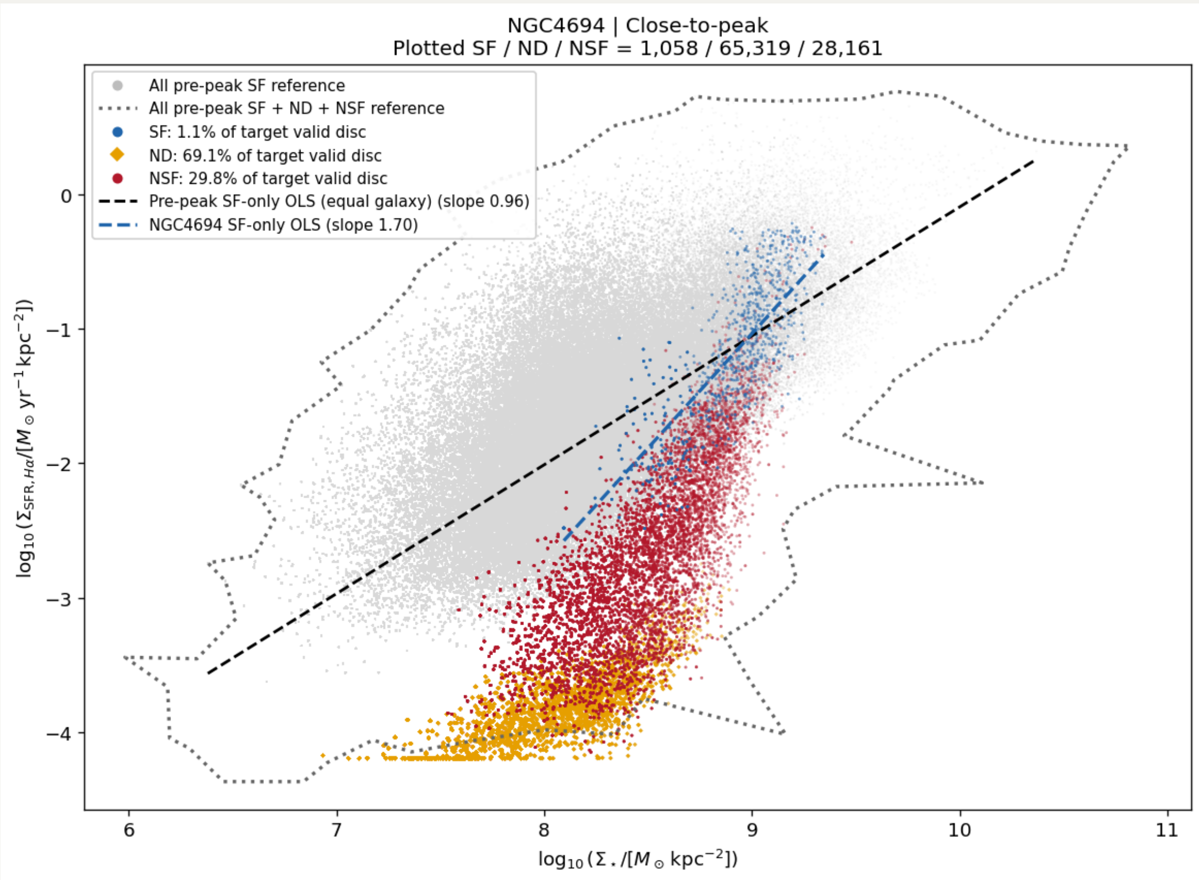


1.5 NGC4694

All looks fine except NGC4694, which i feel like it behaves more like a post-peak. I look at the VIVA paper and it turns out to be a peculiar system with a strange HI morphology and interacting with its neighbour. Anyway, those ND are really NSF.



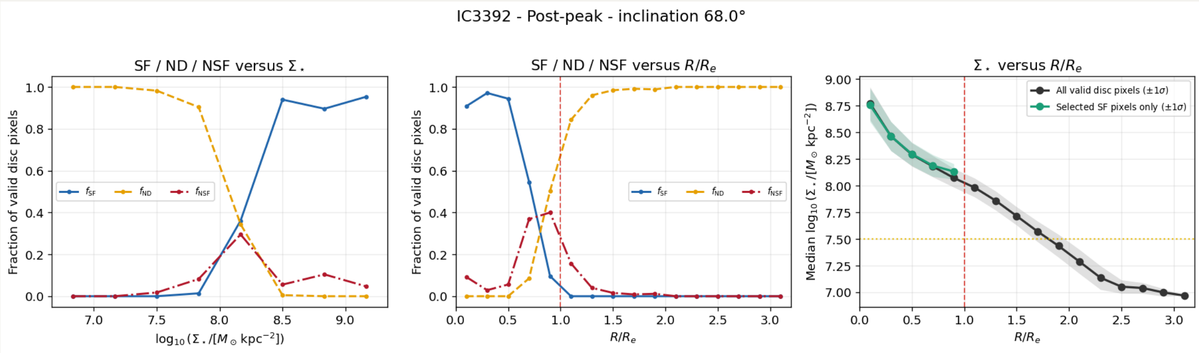


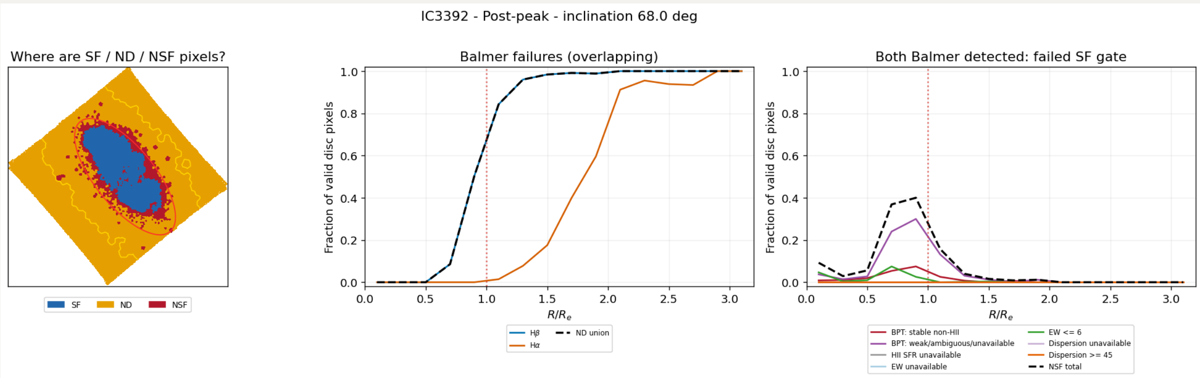


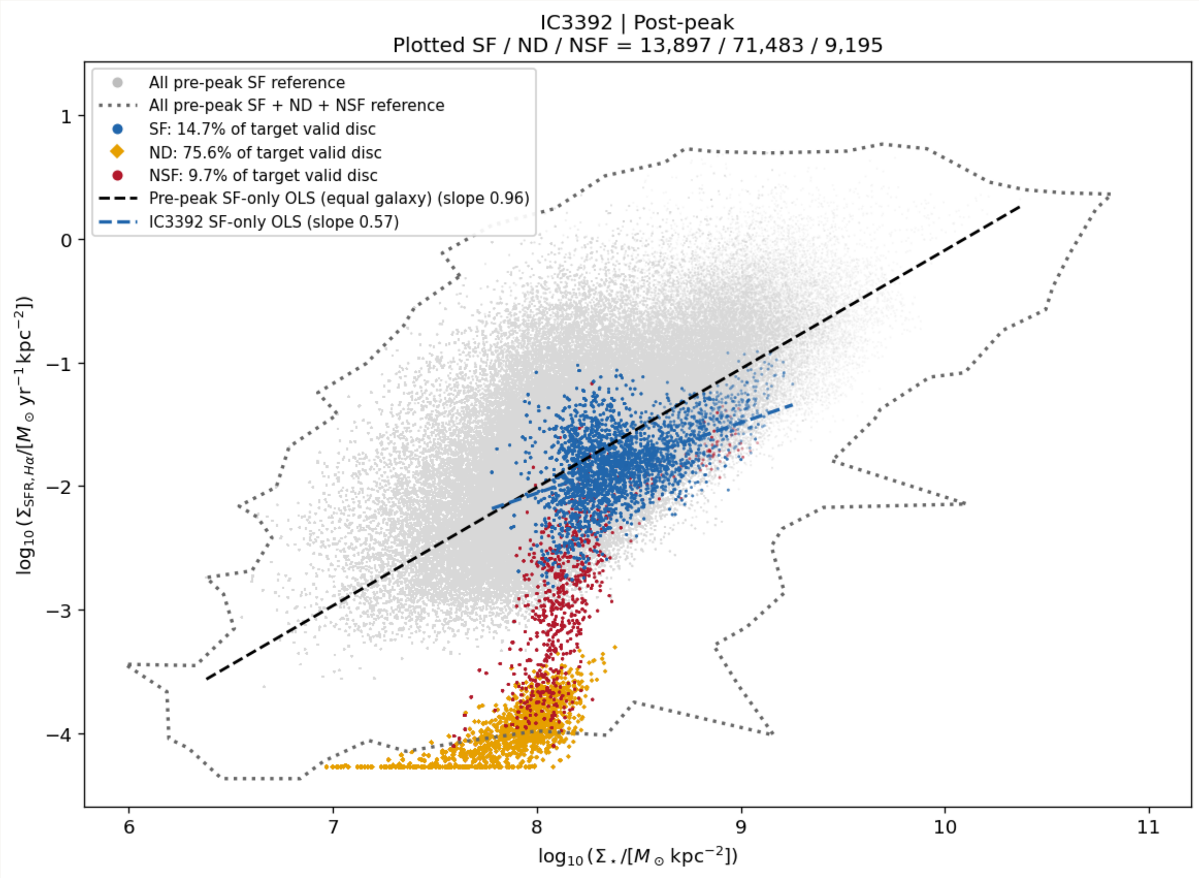
2. Post-peak

2.1 IC3392

Though NSF regions are quite missing with ND, I think those ND regions are no way to be SF.

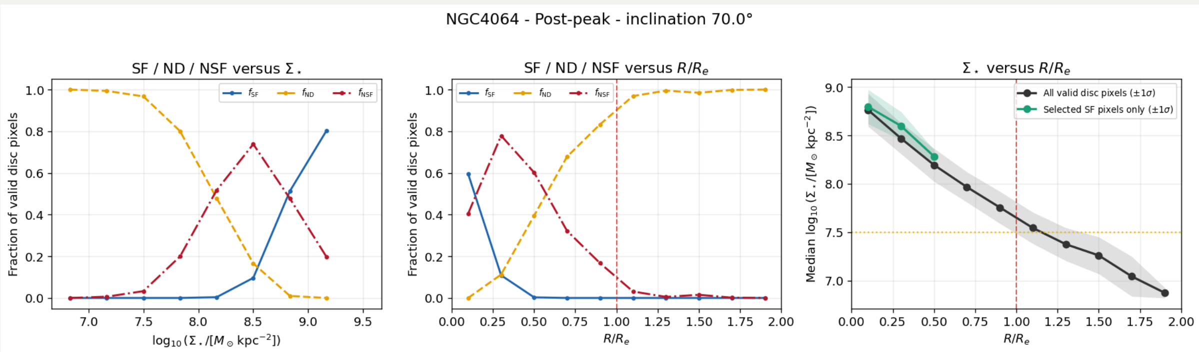


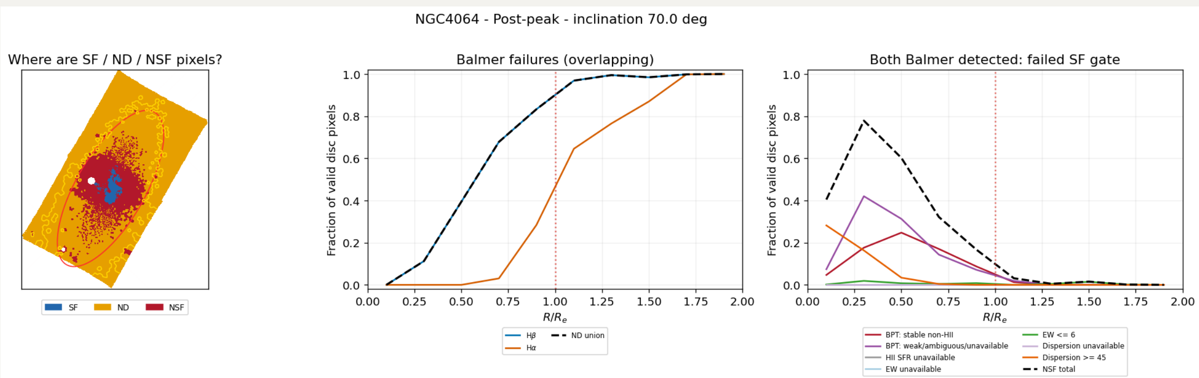


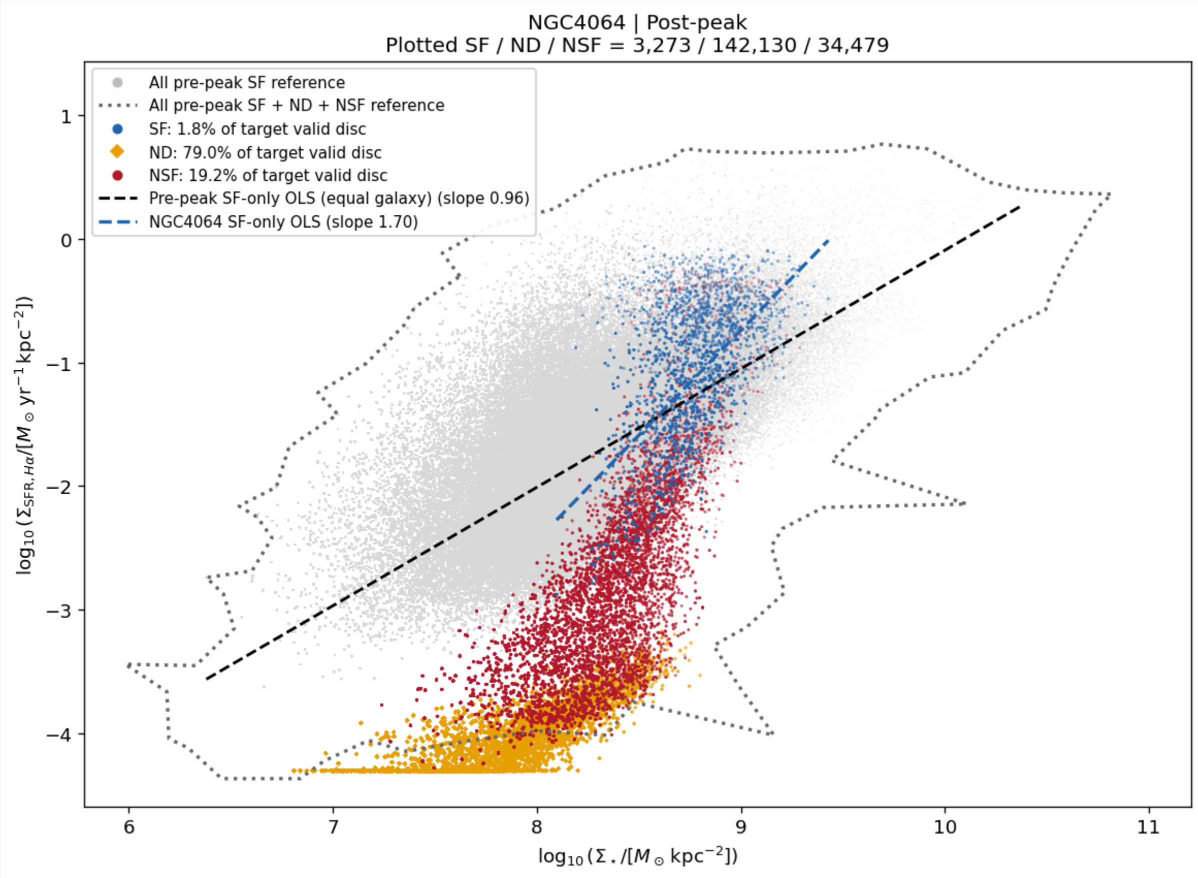


2.2 NGC4064

ND clearly NSF. Σ_{SFR} so high, consistent with its outflow feature.

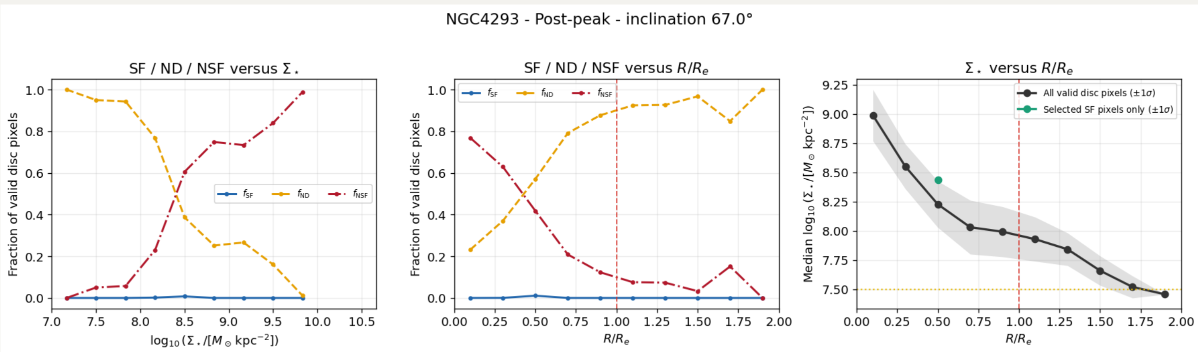


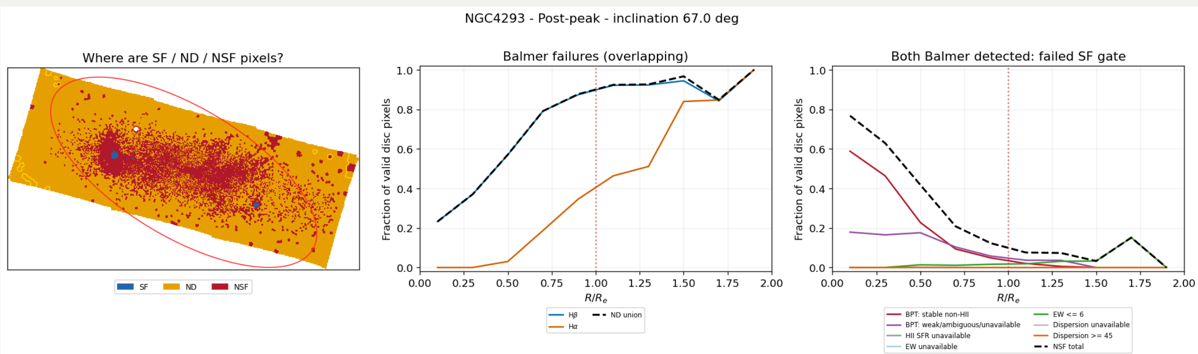


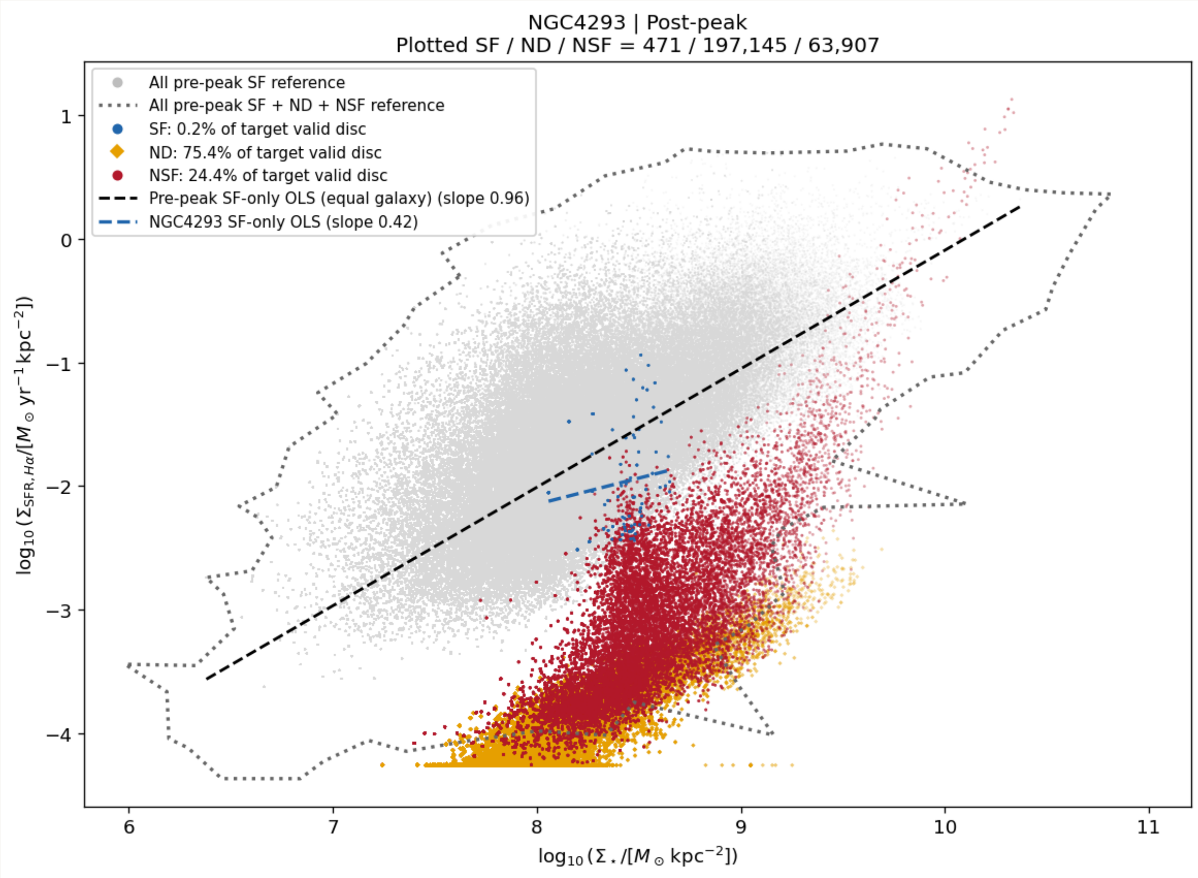


2.3 NGC4293

Man, what can I say, NGC4293 out.

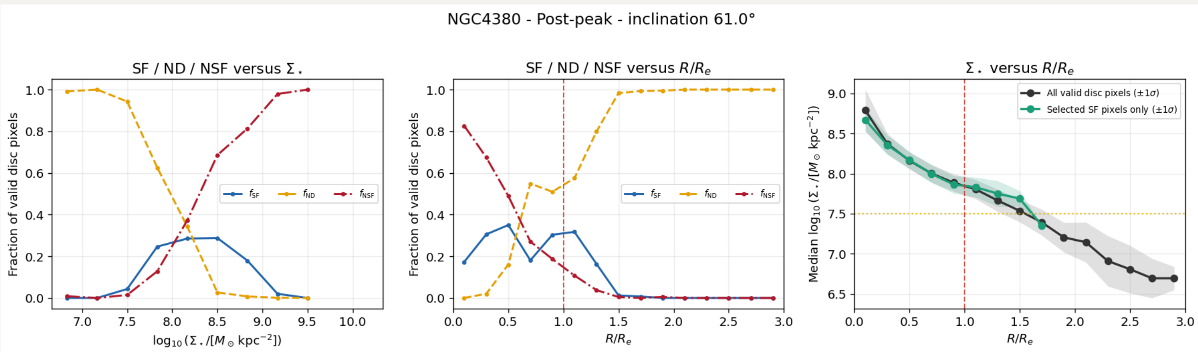


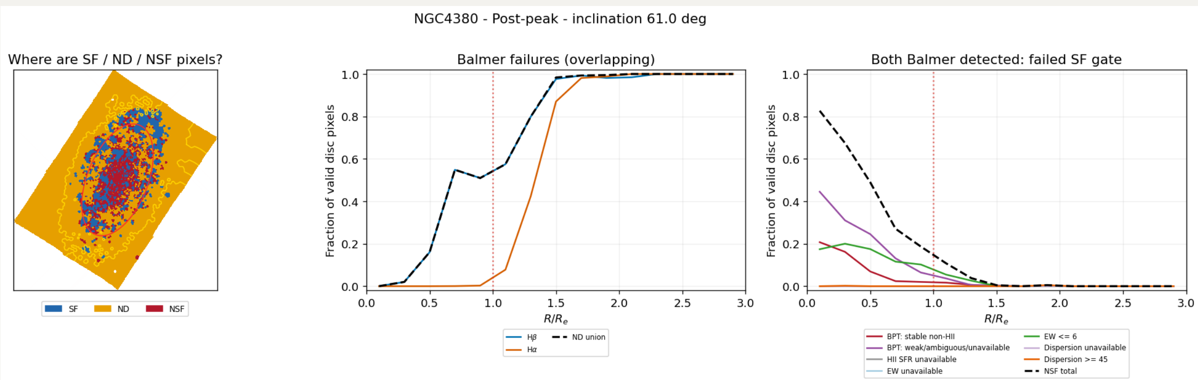


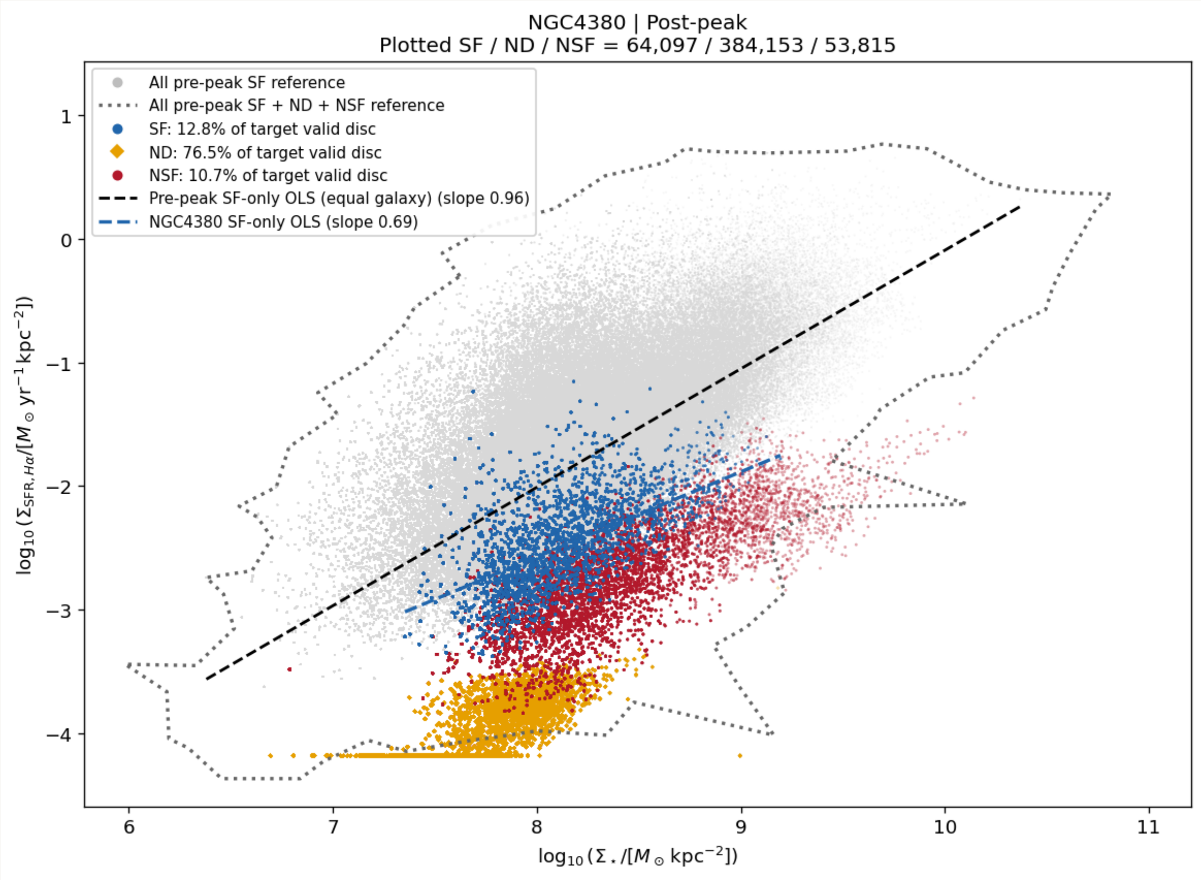


2.4 NGC4380

This one may still have some undetected SF regions.

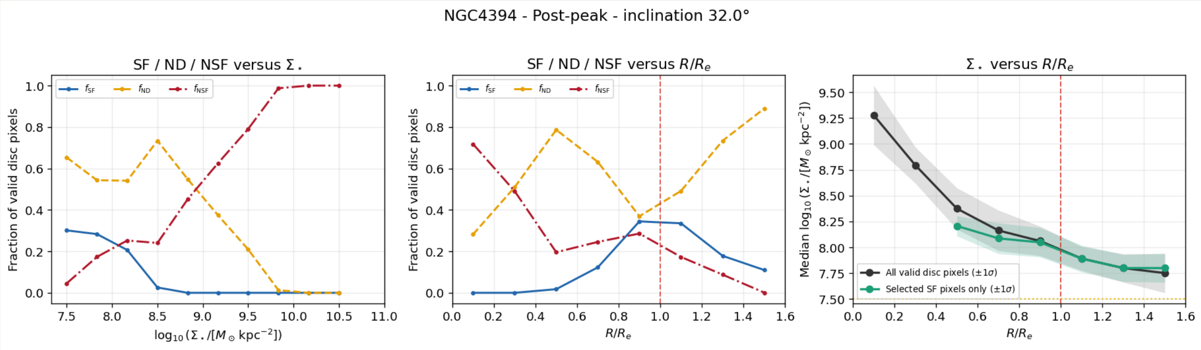


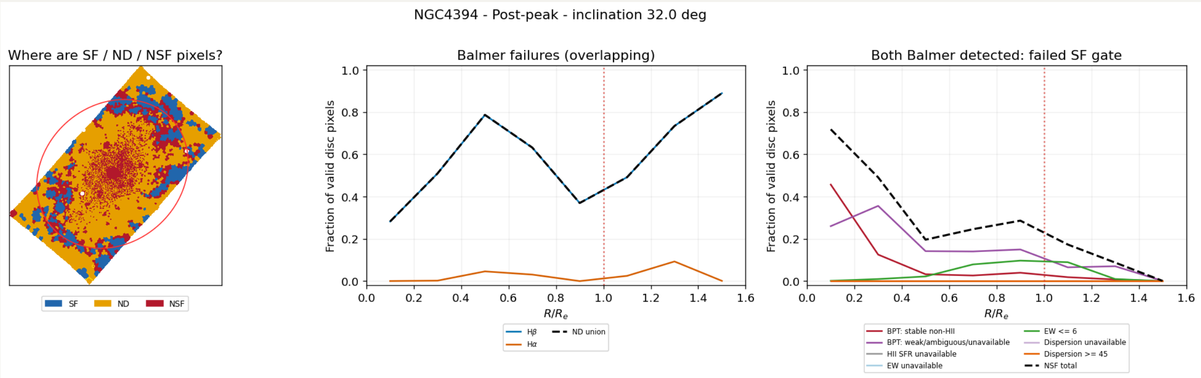


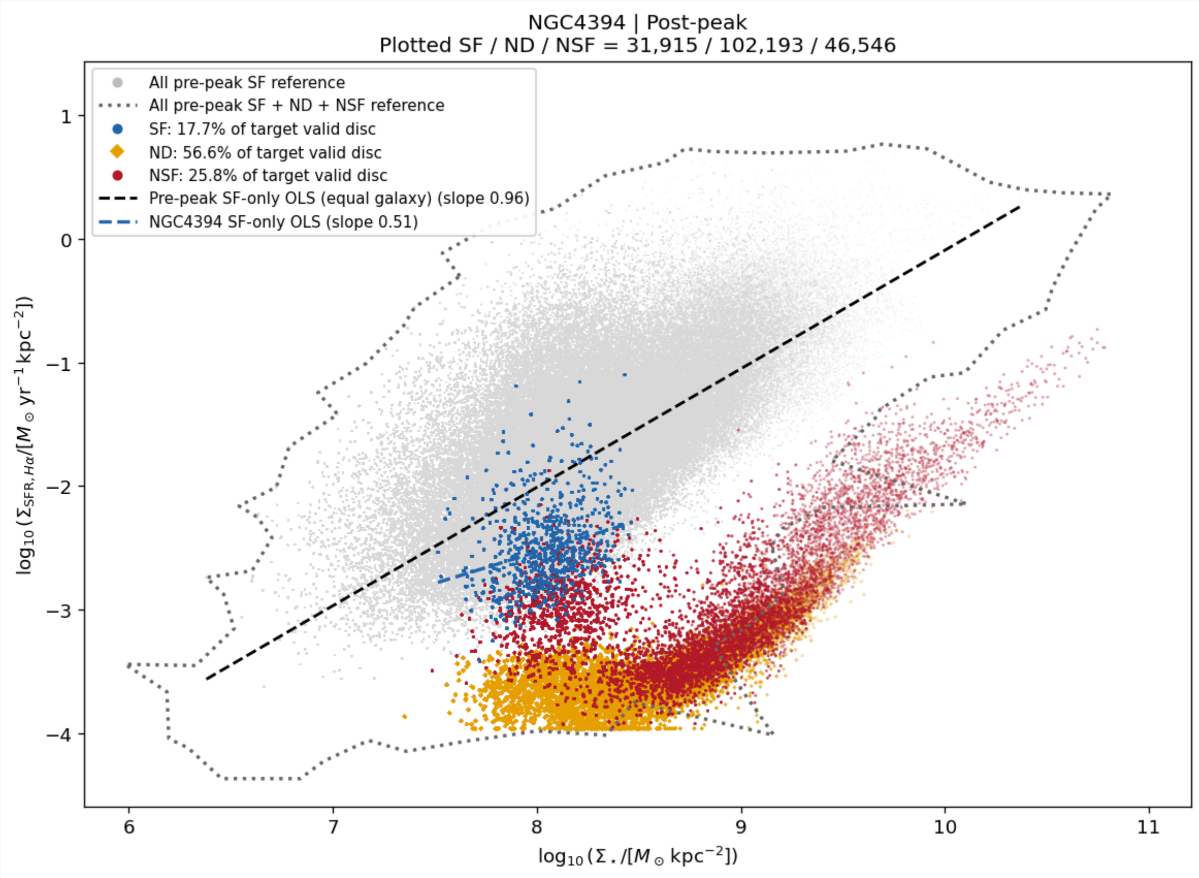


2.5 NGC4394

A suppressed SF ring is presented and may still have some undetected SF regions.

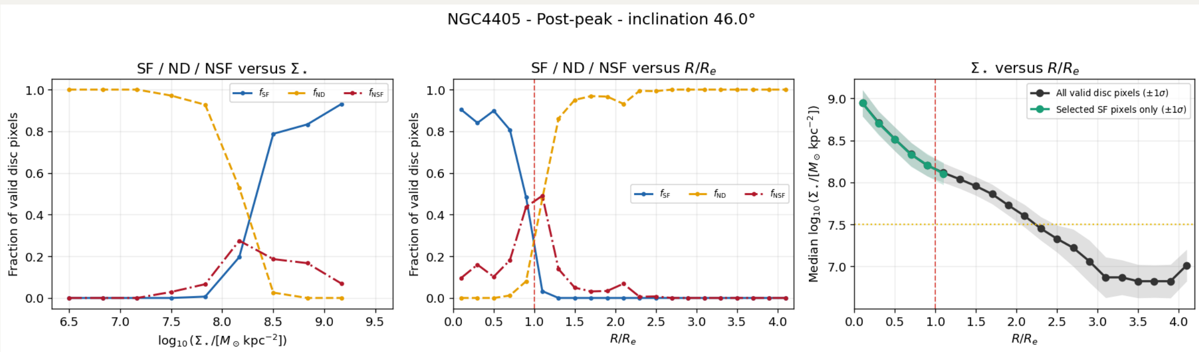


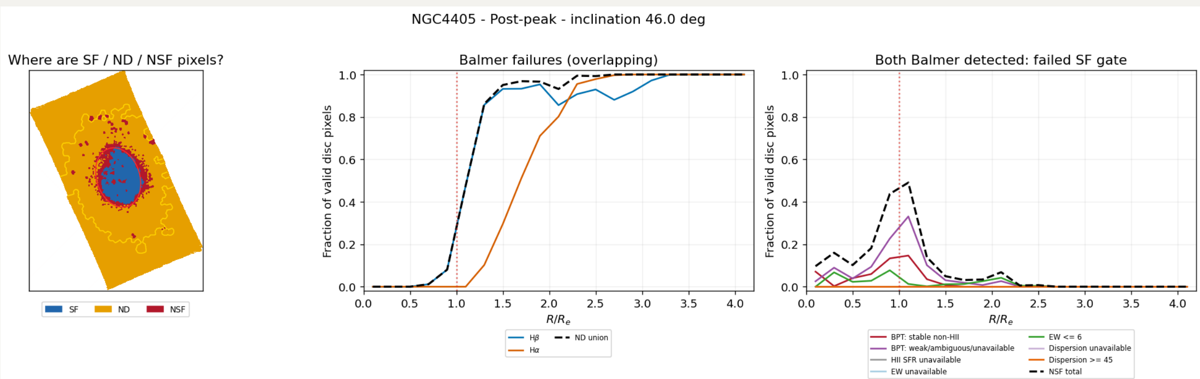


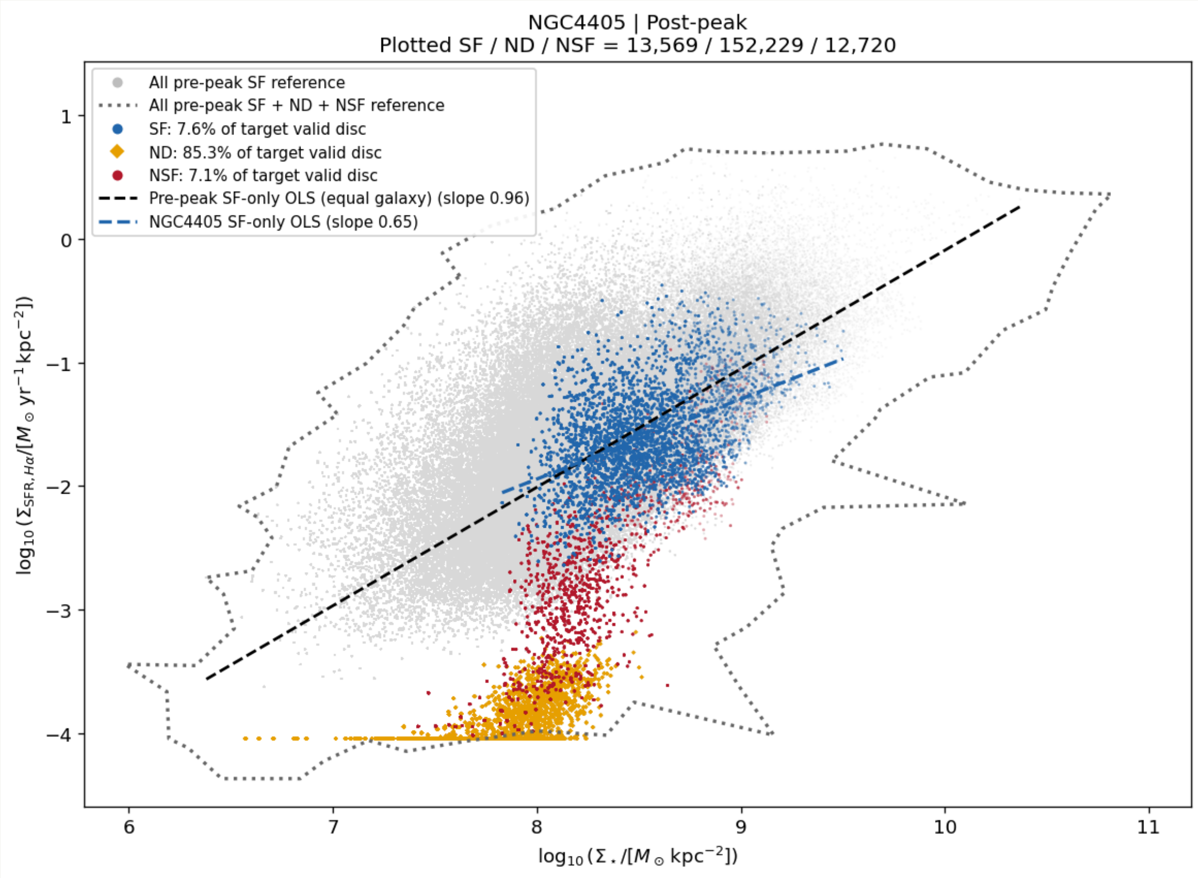


2.6 NGC4405

SF regions are concentrated in the center and surely no potential SF regions in ND.

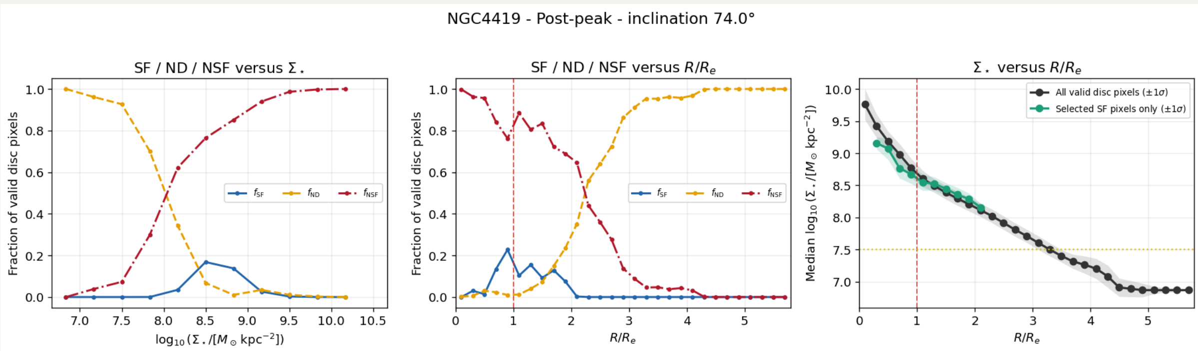


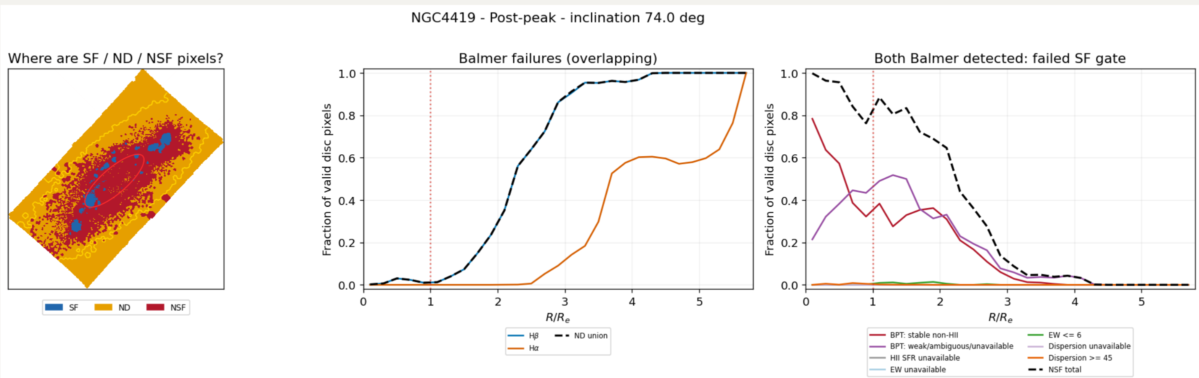


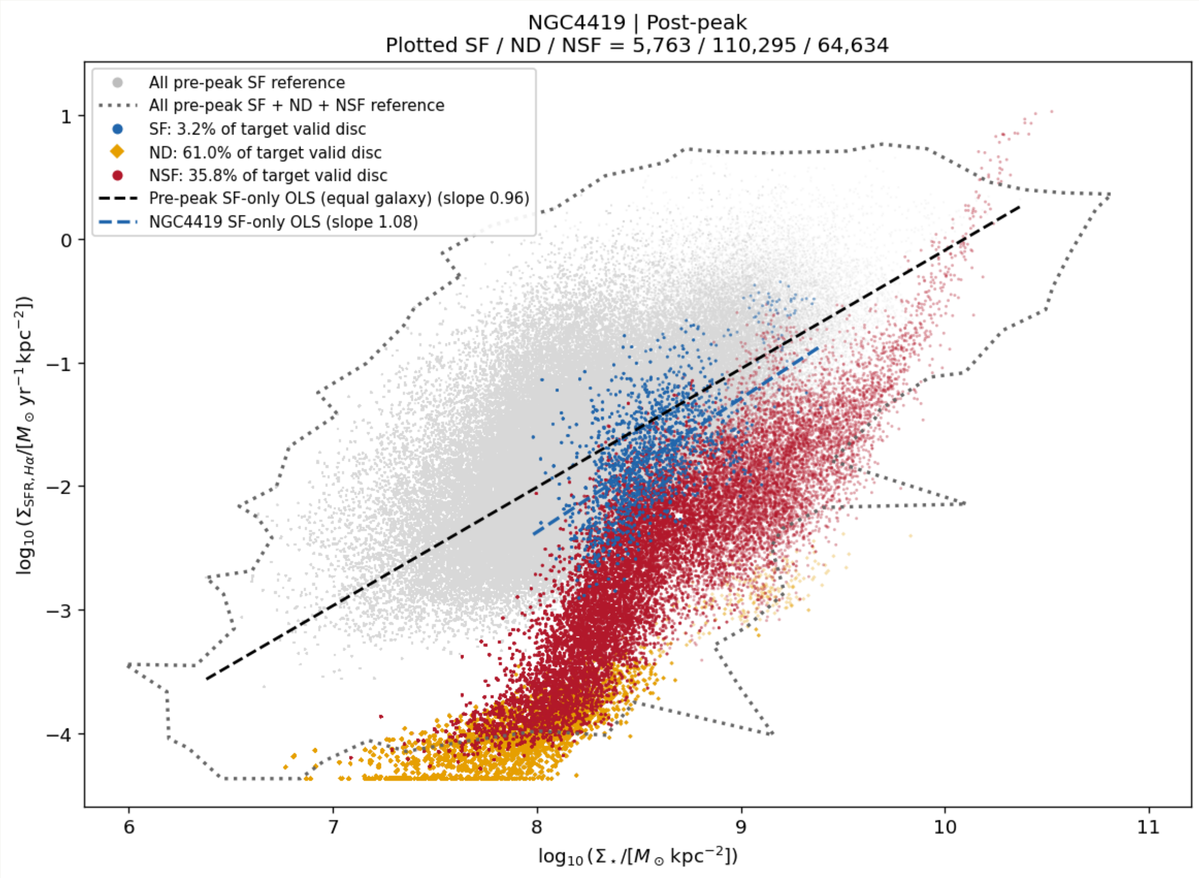


2.7 NGC4419

Only SF regions in a few intermediate mass areas and Σ_{SFR} are suppressed. Also, I will not say there are still some SF regions in those outer ND regions.

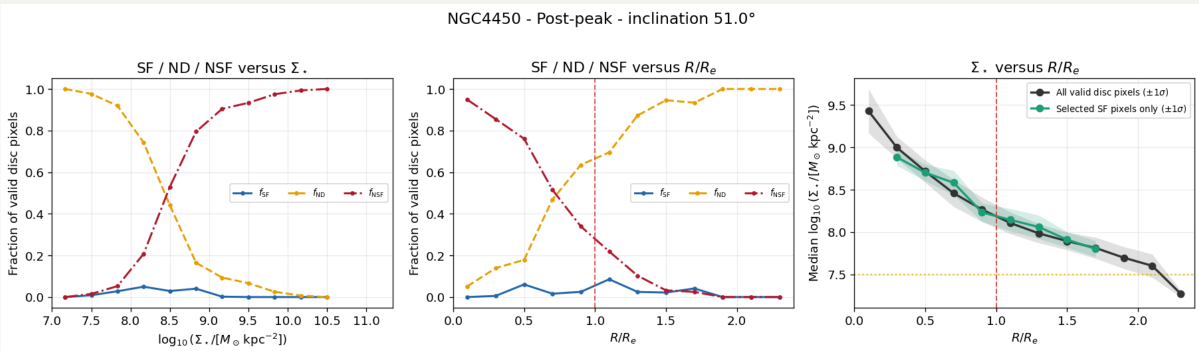


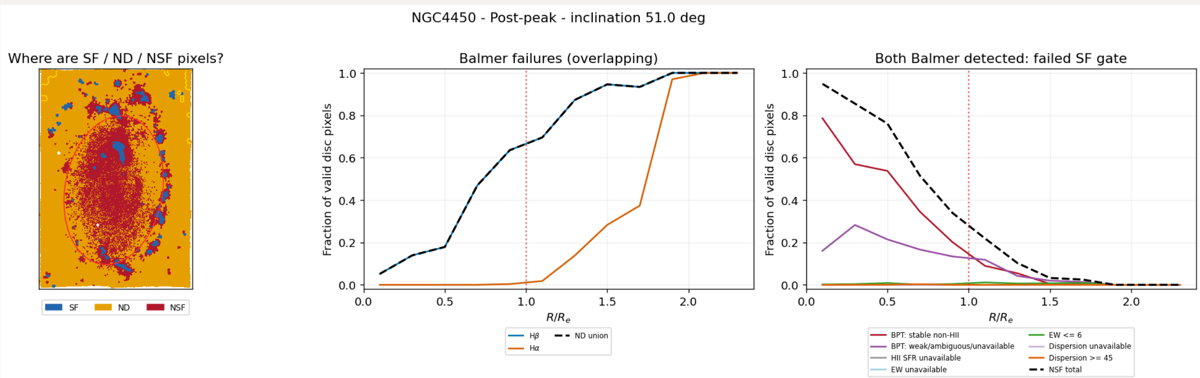


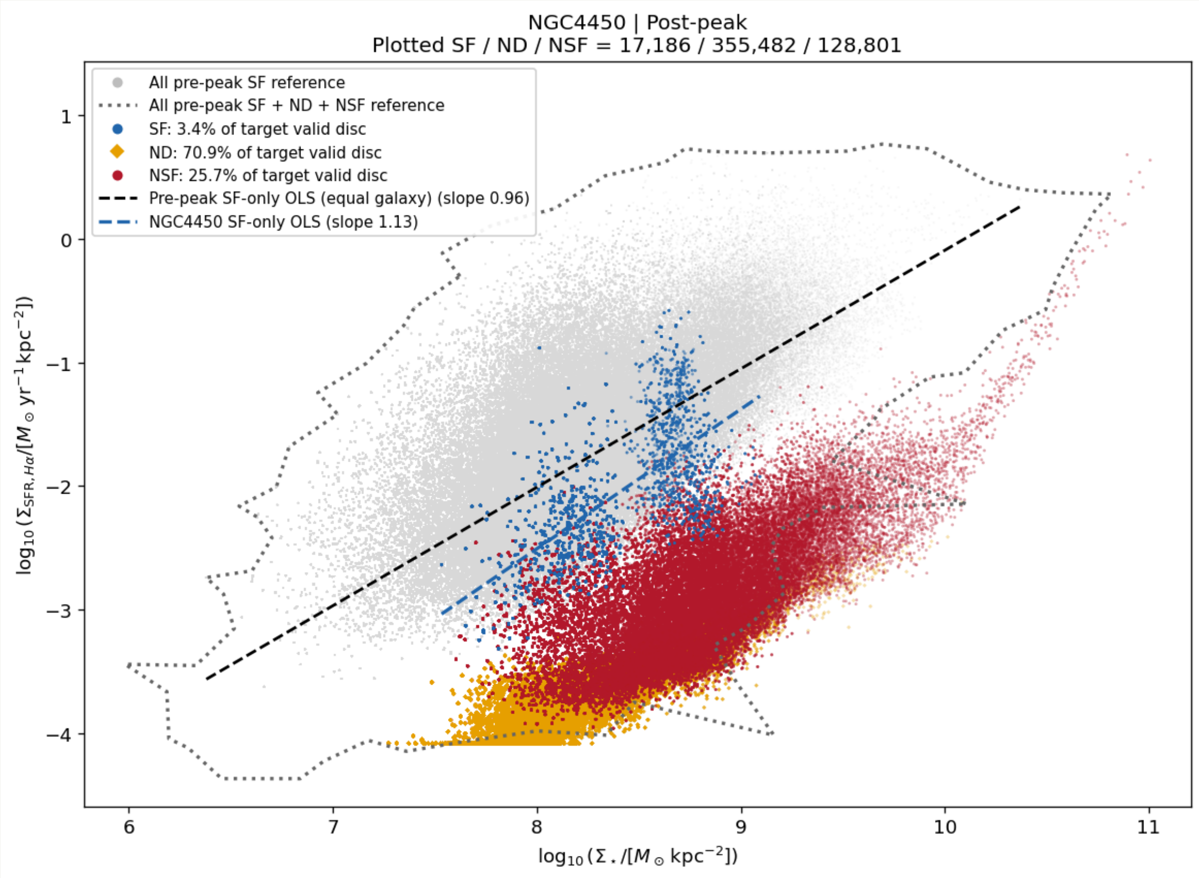


2.8 NGC4450

Suppressed SF regions are dustributed only on the "broken" spiral arms. ND should be NSF.

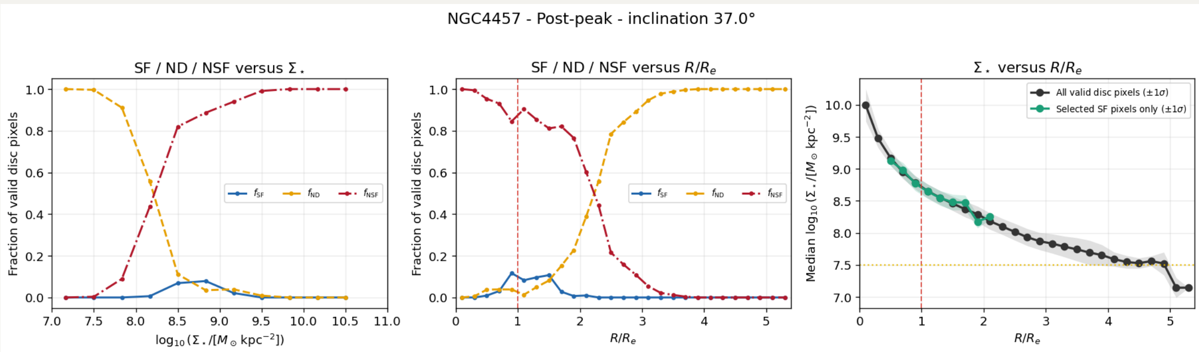


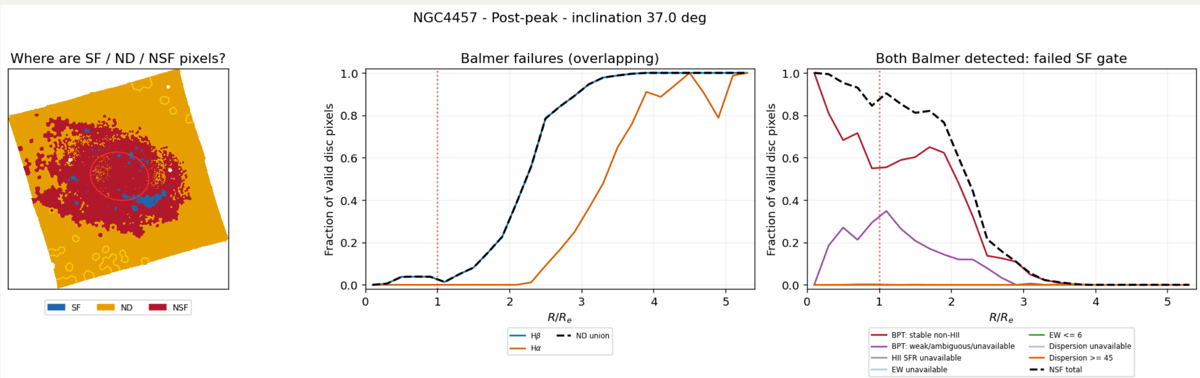


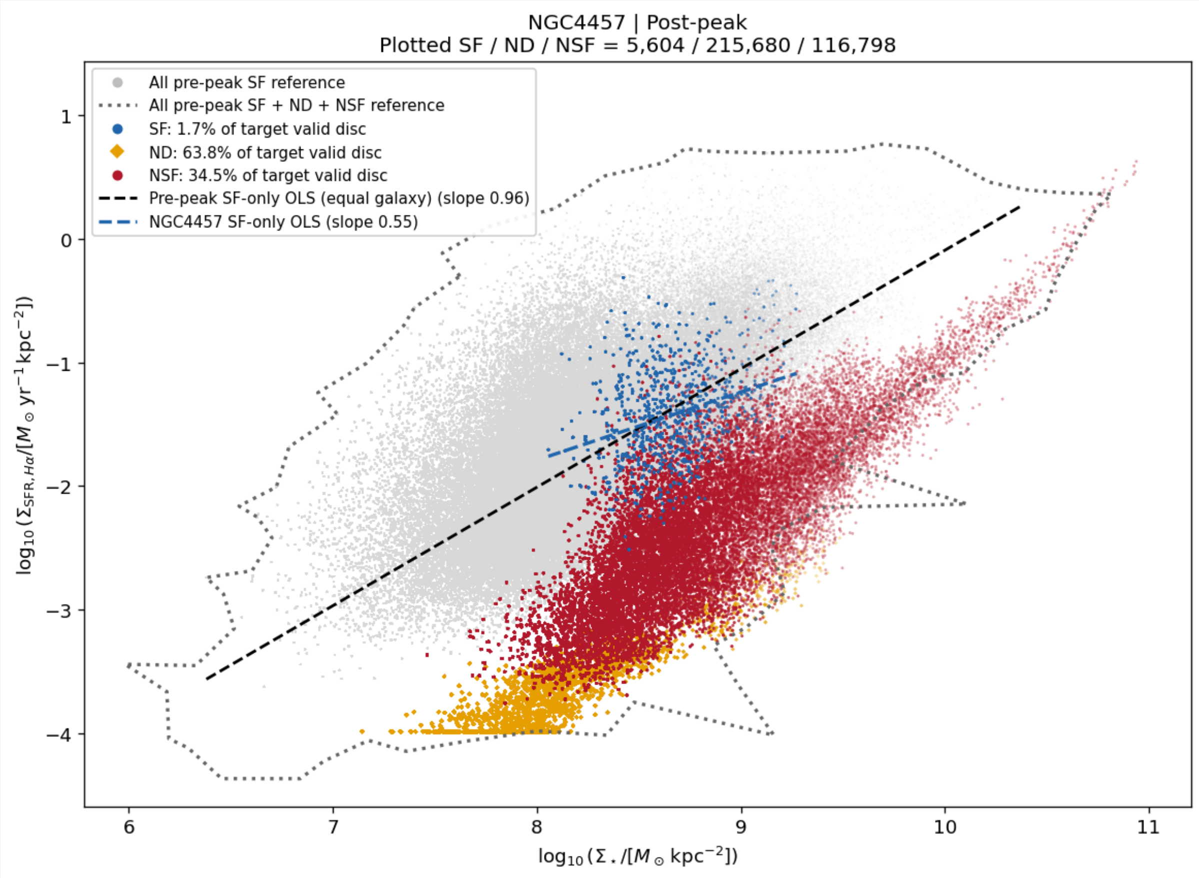


2.9 NGC4457

Though only a few SF spaxels, they look still roughly "normal" on rSFMS. ND should be NSF.

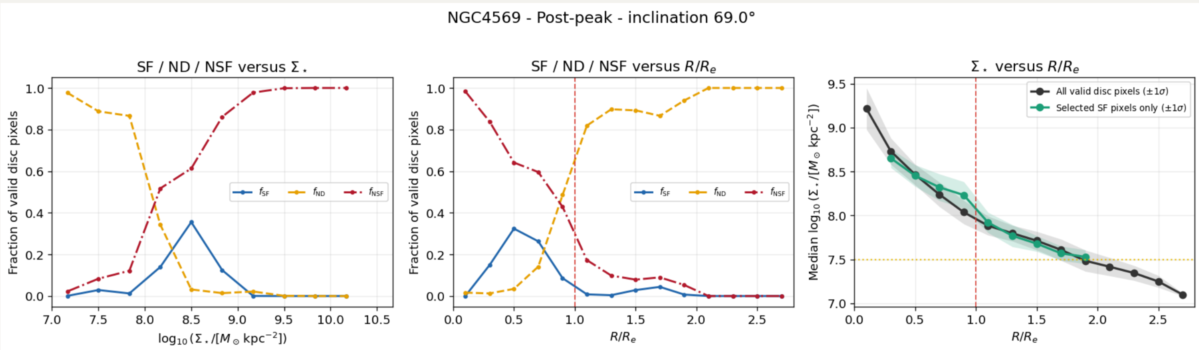


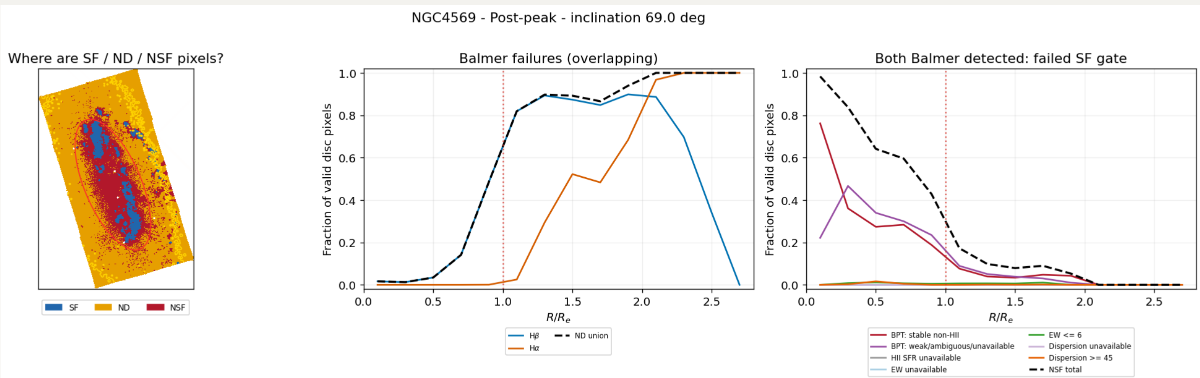


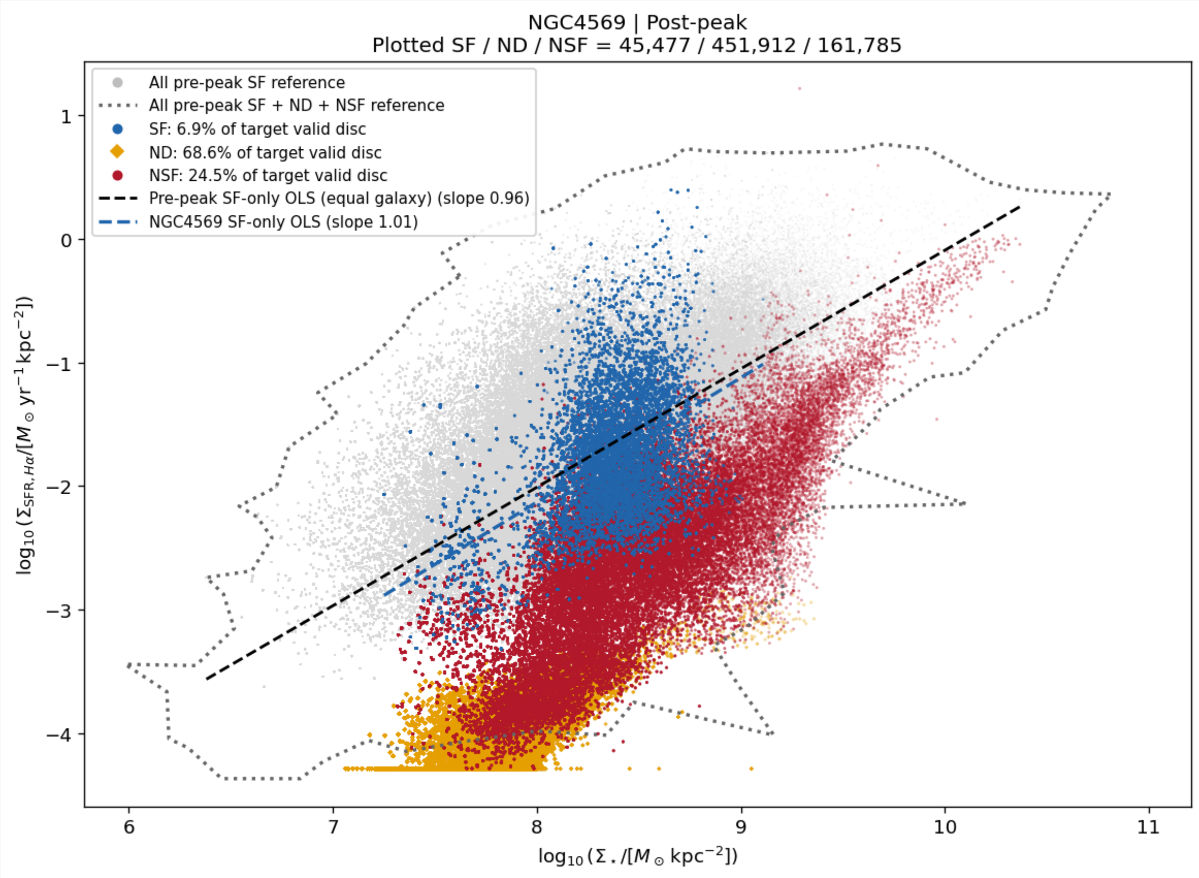


2.10 NGC4569

Again, it seems to another one with "broken" spiral arms, but its rSFMS looks normal. No potential SF regions should be considered.

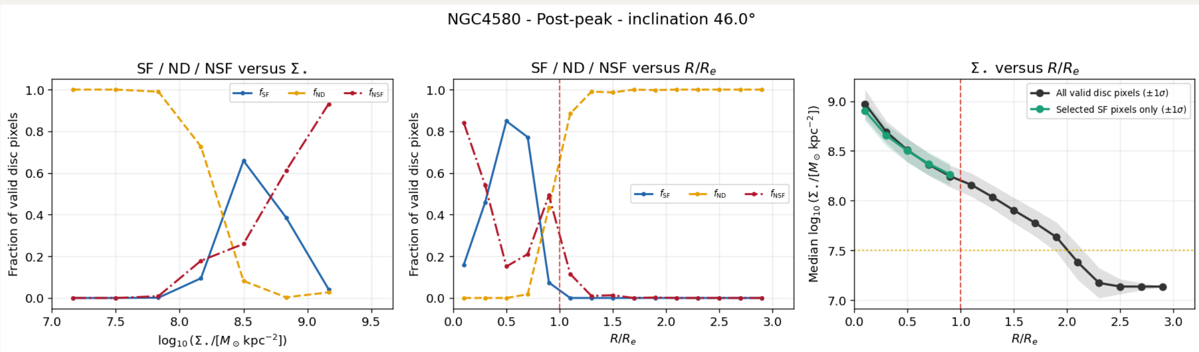


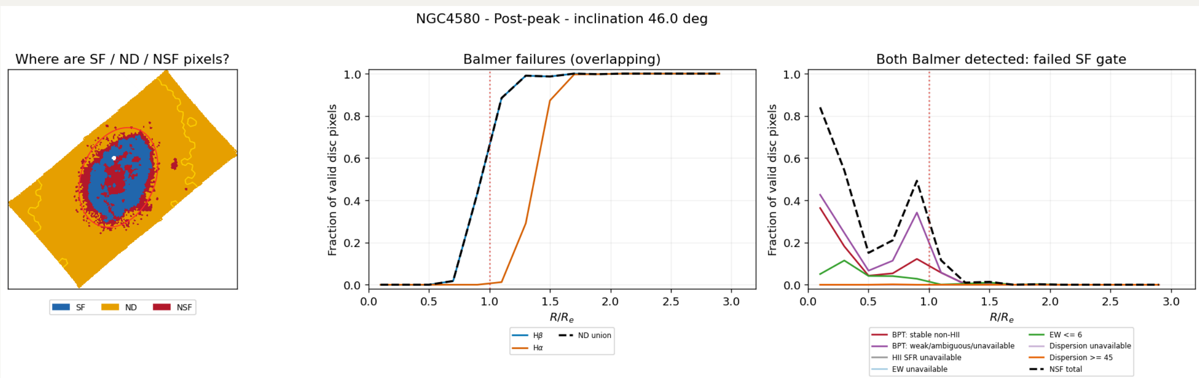


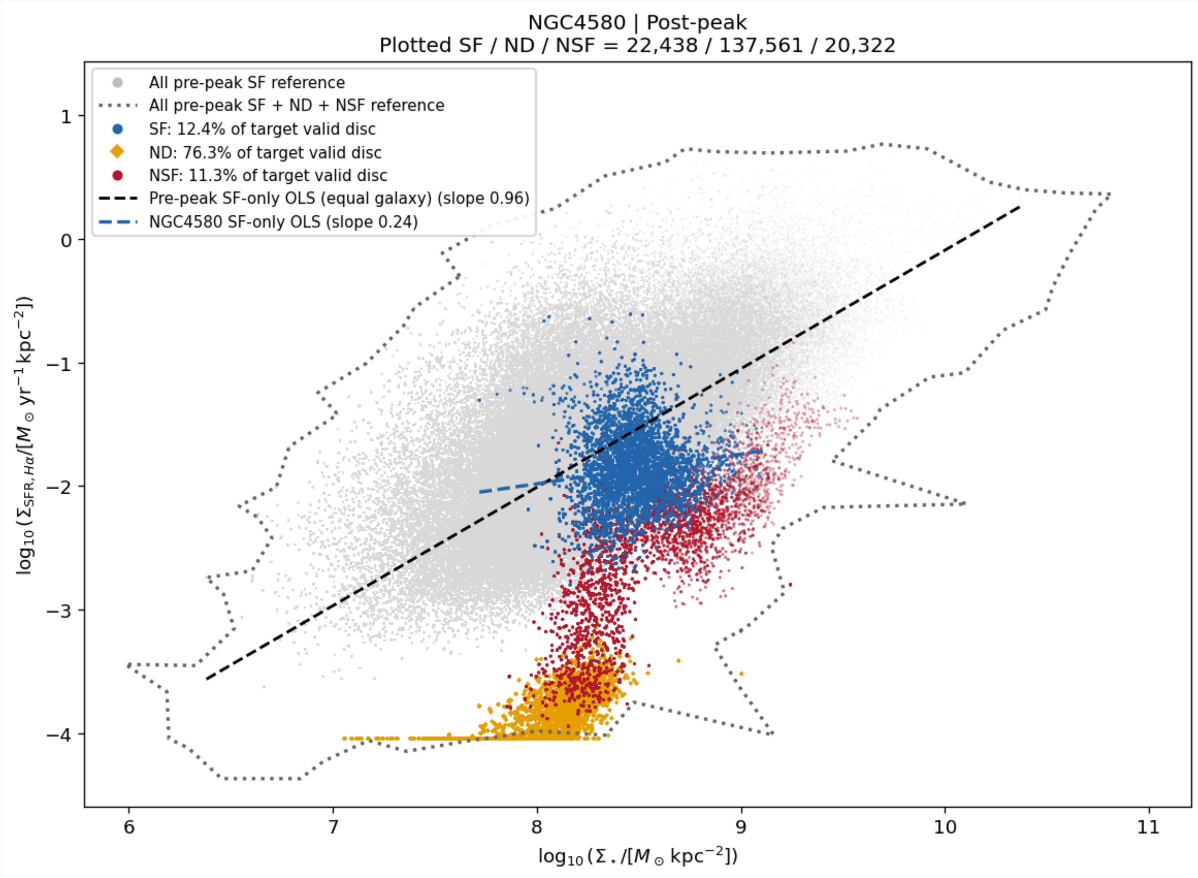


2.11 NGC4580

A bit suppressed SF ring in the center. ND are NSF.

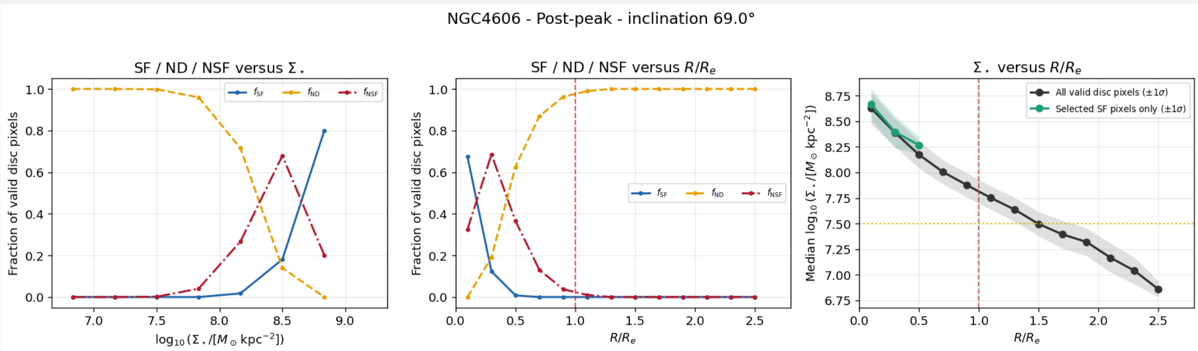


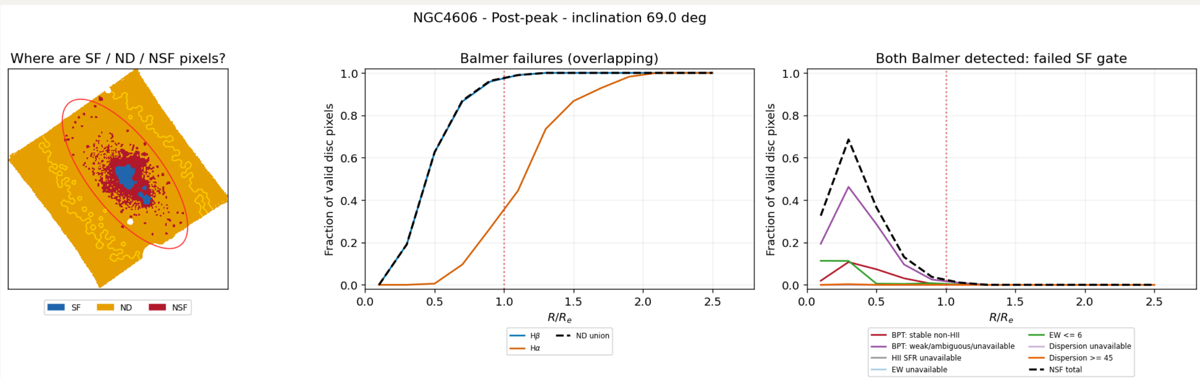


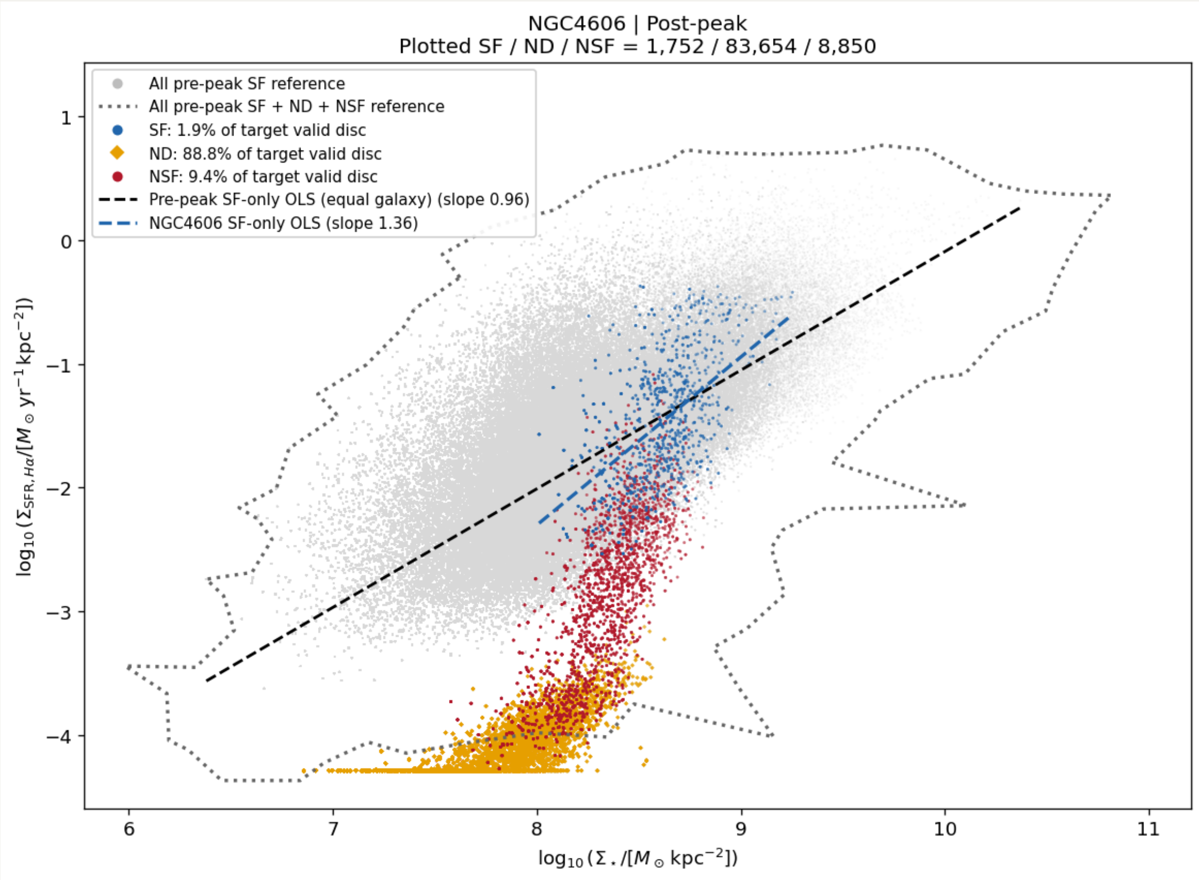


2.12 NGC4606

3 distinct SF regions in the center, surrounded by NSF, so ND are NSF. The rSFMS looks normal, even with some high $\Sigma_{\rm SFR}$ in high mass end, i.e., the galactic center.

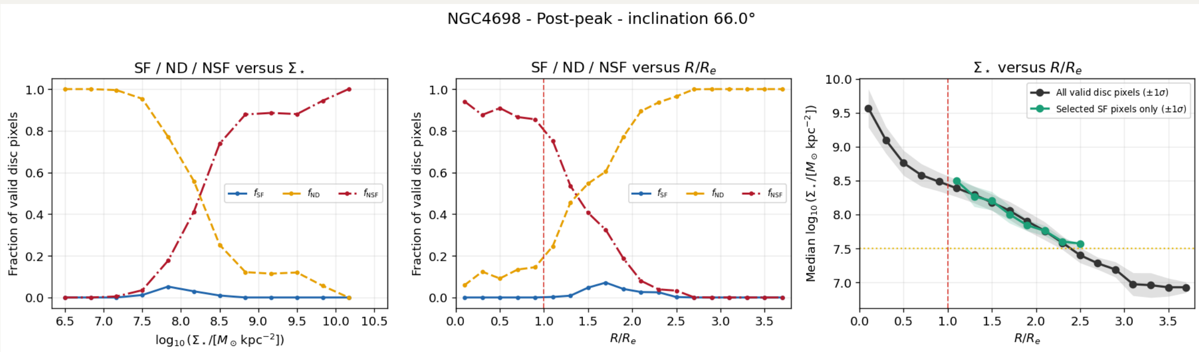


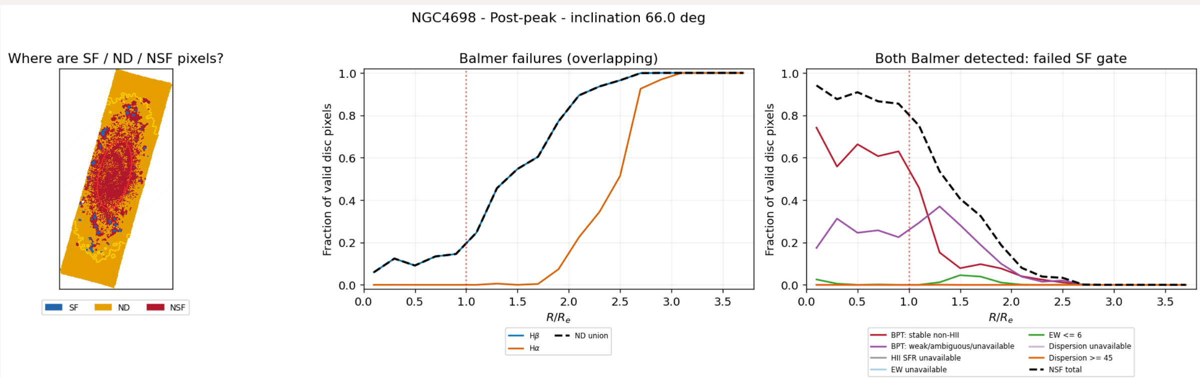


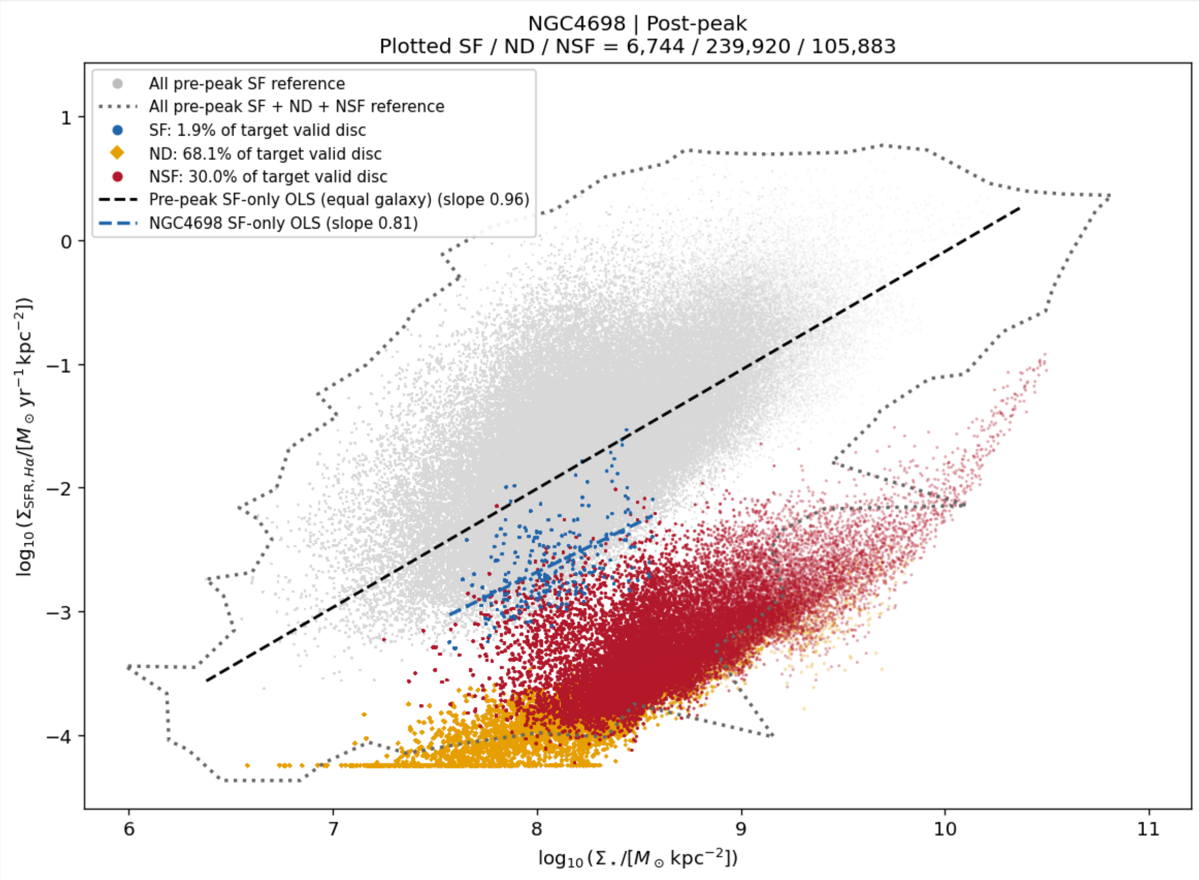


3.13 NGC4698

Again, another suppressed SF region, and even broken. This may suggest that it is accreting some new gas to triger SF in that stellar ring region. No way to have SF in ND.






NGC4698 | Post-peak
Plotted SF / ND / NSF = 6,744 / 239,920 / 105,883
All pre-peak SF reference
All pre-peak SF + ND + NSF reference
SF: 1.9% of target valid disc
ND: 68.1% of target valid disc
NSF: 30.0% of target valid disc
Pre-peak SF-only OLS (equal galaxy) (slope 0.96)
NGC4698 SF-only OLS (slope 0.81)
$\log_{10}(\Sigma_{\mathrm{SFR},H\alpha}/[M_\odot\,\mathrm{yr}^{-1}\,\mathrm{kpc}^{-2}])$
$\log_{10}(\Sigma_\star/[M_\odot\,\mathrm{kpc}^{-2}])$